Decoding Minoan writing (VERSION 3.0.7)

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"... What I would like to share with you today here in Crete is our reading of the (Phaistos) disc and our odyssey of understanding in the linguistic labyrinth on our way to our own Ithaca at Phaistos, Messara, south Crete. It was necessary to have collaboration, constructive criticism, and correction in the learning process. Sometimes it is indeed necessary to take a step back in order to progress in the labyrinth of life, do not be afraid to go back and try again. There is always a better way if we want to find it. There is a win-win scenario. What we must do is to dare to try, to listen, to work together and don't be afraid in life to make mistakes. It took me a long time to realize that. Don't be afraid in life to make mistakes, but to still go ahead preferably with a friend by your side. As we do not yet have a Minoan Rosetta stone, nor the genius of Ventris who deciphered Linear B, and as we didn't get the Nobel for literature, then we must keep trying..."

(Gareth Owens, TED Talks: TEDxHeraklion, May 23rd, 2014)
(https://www.youtube.com/watch?v=6Chcplx3tZ8)

(B2, Phaistos Disc, ca. 1,600 BC)

CONTENT

PRESENTATION 7
CHAPTER I. THE PHAISTOS DISC 9
INTRODUCTION 11
RULES FOR READING 12
PRELIMINARY CONSIDERATION 12
PROPOSED MEANING OF SYMBOLS 13
READING THE SIDE "A" OF THE DISC 17
READING THE SIDE "B" OF THE DISC 20
PRAYER (CONCEPTUAL) FROM THE PHAISTOS DISC 22
COMMENT 24
ABOUT CONSENSUS 24
ABOUT MINOAN POTTERY AT PHAISTOS 27
CHAPTER II. AXE OF ARKALOCHORI 31
INTRODUCTION 33
READING THE SYMBOLS OF THE AXE OF ARKALOCHORI 33
MESSAGE (CONCEPTUAL) OF THE AXE OF ARKALOCHORI 34
COMMENT 35
ABOUT CONSENSUS 36
PARTIAL PARALLELISMS OF TEXTS IN ARKALOCHORI 37
CHAPTER III. ALTAR OF MALIA 41
INTRODUCTION 43
READING THE SYMBOLS OF THE ALTAR OF MALIA 43
MESSAGE (CONCEPTUAL) OF THE ALTAR OF MALIA 44
COMMENT 44
ABOUT CONSENSUS 45
CHAPTER IV. FRAGMENT HM_992 47
INTRODUCTION 49
READING THE SYMBOLS OF THE FRAGMENT HM_992 49
COMMENT 49
CHAPTER V. CONCLUSION 51
THE MINOAN WRITING IS IDEOGRAPHIC 53
THE MINOAN WRITING HAS BEEN DECIPHERED 56

ABOUT THE CRETAN HIEROGLYPH 57
ABOUT THE MINOAN LINEAR A 58
APPENDICES 61
APPENDIX A. DISCUSSION ON THE NEW MINOAN SYMBOL 63
APPENDIX B. REDISCOVERING AN ANCIENT STAR 65
APPENDIX C. MORE CERAMICS AND THE VENUS OF BENAOJÁN 71
APPENDIX D. TABLE OF MINOAN SYMBOLS CORRESPONDENCE 73

PRESENTATION

The present work aims to decipher the Minoan writing system, associated with the Minoan civilization that inhabited the island of Crete during the Bronze Age.

The following table[1] shows the different writing systems that appeared on the island of Crete and mainland Greece during the Bronze Age. It is believed that the change from one to the other was not abrupt, but that there were periods of overlap and coexistence.

Writing System	Geographical Area	Approx. time span
Cretan hieroglyph (unknown)	Crete (east of the Knossos-Phaistos axis)	2,100–1,700 BC
Minoan Lineal A (unknown, syllabic??)	Crete (except the southwestern tip), Aegean Islands (Kea, Kythera, Melos, Thera), and mainland Greece (Laconia)	1,800–1,450 B.C.
Mycenaean Linear B (syllabic, deciphered)	Crete (Knossos), and mainland Greece (Pylos, Mycenae, Thebes, Tiryns, Agios Vasileios – the ancient name of the latter is unknown)	1,450–1,200 BC
Cypro-Minoan	Cyprus	1,550–1,050 BC
Cypriot	Cyprus	11th–4th centuries BC

The inscriptions in **Cretan Hieroglyph** can be divided into 3 basic categories[2]:

1. Clay tablets and sealings.
2. Sealstones.
3. Other inscriptions, namely the "stone Altar of Malia", the "Axe of Arkalochori" and the "Phaistos Disc"[3].

The set of these "other inscriptions" (and in the so-called Fragment HM_992) is commonly **known as the Minoan script** and, as the whole Cretan Hieroglyphics, it remains undeciphered. Minoan Linear A script is also undeciphered, whereas Mycenaean Linear B was deciphered in 1952 by British architect and palaeographer Michael Ventris OBE.

This paper will explore the new approach of considering every symbol of this Minoan script as an **ideogram**, instead of the syllabic approach normally preferred by the specialists.

We are going to study the four known archaeological artefacts that are officially recognized to contain this Minoan script:

- ✓ The Phaistos Disc.
- ✓ Axe of Arkalochori.
- ✓ Altar of Malia.
- ✓ Fragment HM_992.

[1] Table taken from the Wikipedia website. Link: (https://en.wikipedia.org/wiki/Cretan_hieroglyphs).

[2] Gareth Owens (1996), "Evidence for the Minoan Language (5)-The Phaistos Disk". Taken from web Daidalika. Link: (https://daidalika.hmu.gr/minoan_scripts-bibliography/Ph.D. 1 (1996)).

[3] The artefact is often referred as the Phaistos "Disk", instead of "Disc", perhaps due historical reasons. Instead, when they refer to its elements, some use "disk" and some others "disc". As there is not any semantic argue in favour of the use of "disk", in this study we will only use the more appropriate term of "disc".

The key points developed in this document are as follows:

- The pure ideographic approach is perfectly valid to approach the project of trying to decipher the so-called Minoan language, a subset of the Cretan Hieroglyph.

- The result we obtain using this approach is spectacular: the translation is of great beauty and perfectly coherent both with each and every one of the artifacts studied (Phaistos Disc, Axe of Arkalochori and the Altar of Malia) and with the context we know of the Minoan people. This result strongly suggests that the Minoan script may be of a pure ideographic character.

- At the same time, we present a good number of arguments against the syllabic approach, preferred by most researchers, using as a common thread the work of the team of Gareth Owens (Associate Director, Hellenic Mediterranean University), a renowned researcher and expert in this field.

- In particular, we strongly refute the argument that the Minoan script can be of a syllabic type only because it is a predecessor (not a precursor) very distant from the Mycenaean Linear B, a syllabic language that appeared on the island of Crete several centuries after any form of Cretan hieroglyph and came from the hand of a people, the Mycenaeans, who subdued and put an end to the Minoan civilization.

- We present evidence that the later Minoan Linear A language could be, at least in its beginnings, also of a pure ideographic character.

- Specifically, and with respect to the Phaistos Disc, we have identified 46 different symbols, compared to the 45 that appear in all the documents published on the subject. That is, we have discovered a new Minoan symbol that also has its historical relevance: we show that this new symbol, also identified in the Axe of Arkalochori, would be an early representation of the famous symbol known as "triskelion".

- The Phaistos Disc must be read following the spiral on both sides in one direction from the centre outwards, that is, in the opposite direction to the current consensus.

- We show that the vertical or oblique marks that appear from time to time on the symbols of the disc are "new paragraph" marks, which denotes the use of paragraphs, as had been proposed many decades ago.

- The translation obtained from the Phaistos Disc, which turns out to be a beautiful communal prayer to their gods, suggests that it may have been created after a great natural catastrophe, which could well be the collapse of the volcano on the neighbouring island of Santorini around 1,600 B.C.

- The formulas identified in the Phaistos Disc are phrases that are still used today in the prayers of Christianity, indicating the antiquity of these formulas and how they have been passed down throughout history from civilization to civilization.

- From the relationship found with other archaeological artifacts found in locations in present-day Turkmenistan and Pakistan, we can see how at an initial moment in the development of languages, the appearance of ideograms allowed for the first time to disseminate ideas and concepts shared between different peoples in written form.

CHAPTER I. THE PHAISTOS DISC

INTRODUCTION

It is a clay disc of about 16 cm in diameter, discovered in the Palace of Phaistos, on the island of Crete, in 1908, attributed to the Minoan civilization, with an imprecise dating between 1,800 and 1,400 B.C. It is on display at the Archaeological Museum of Heraklion, Crete. The consensus says this is the **largest surviving example of the so-called Minoan script**.

On the disc there are a series of symbols on both sides arranged in a row, following a spiral, occupying the entire surface. Each symbol that you see is the result of marking the stamp of each figure one by one on the clay surface, before firing it and hardening it. Some stamps were used several times during the process, as the same symbol is printed several times.

[4]

[5]

Some other archaeological elements have been discovered on the same island of Crete that contain symbols like those on the disc. On the one hand, these might constitute strong evidence about the authenticity of the disc; and on the other hand, indicates that the use of this script was widespread in this island for a given period: as we will see, perhaps not in a generalized way but at least in some specific scope. In these other artefacts the symbols appear in a much more rudimentary way (Axe of Arkalochori, stone Altar of Malia).

The fact that the symbols are displayed on this disc in such a detailed manner, along with the elaborated design, make it clear that this disc is not an example of regular communication. On the contrary, it is a unique piece, something exceptional. It is a true work of art at the time, both for the conceptual design and its execution.

In our attempt to decipher the disc, we will consider every symbol as an ideogram. And by using this new approach, we will reveal how the two sides of the disc actually show a beautiful prayer to the deities asking for all the typical favours for the community.

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[4] Side A of the disc. We recommend opening this link apart and enlarge it to better follow the reading discussion. Photo taken from Wikipedia licensed under Creative Commons Attribution-Share Alike 4.0 International, link: (https://commons.wikimedia.org/wiki/File:%CE%94%CE%AF%CF%83%CE%BA%CE%BF%CF%82_%CF%84%CE%B7%CF%82_%CE%A6%CE%B1%CE%B9%CF%83%CF%84%CE%BF%CF%8D_%CF%80%CE%BB%CE%B5%CF%85%CF%81%CE%AC_%CE%91_6380.JPG).

[5] Side B of the disc. We recommend opening this link apart and enlarge it to better follow the reading discussion. Photo taken from Wikipedia licensed under Creative Commons Attribution-Share Alike 4.0 International, link: (https://commons.wikimedia.org/wiki/File:%CE%94%CE%AF%CF%83%CE%BA%CE%BF%CF%82_%CF%84%CE%B7%CF%82_%CE%A6%CE%B1%CE%B9%CF%83%CF%84%CE%BF%CF%8D_%CF%80%CE%BB%CE%B5%CF%85%CF%81%CE%AC_B_6381.JPG).

RULES FOR READING

The following are some basic observations about the formal structure of the disc, which, from what can be seen in the information available on the Internet today on this subject, differ quite a bit from the current consensus.

- ✓ Whereas the consensus agrees that the disc reads spirally from the outer border to the centre for both sides in a clockwise orientation, during this study it clearly shows just the opposite: the **symbols should be read from the centre of the disc, following the spiral line, to its outer end[6],** which on both sides is indicated by a row of 5 vertical holes. The spiral runs counterclockwise, always **following the orientation of the human figures, for example**.
- ✓ Whereas the consensus agrees that the disc shows syllables or phonetic elements in the forms of symbols that finally would compose some few words, we state that each of the symbols is either a **pictogram or an ideogram**, that is, it represents an idea, an object, or an action. And it must be interpreted in terms of adjacent symbols, as they modify or complement its meaning.
    - There are symbols that almost always appear in pairs, which would indicate that they are related, they should be read together (e.g. "shield" + "man", "snake" + "hawk").
    - The use of duplicate symbols (e.g. "hide" + "hide", or "palace" + "palace") indicating the use of the plural is observed, although only in the case of nouns, it does not affect adjectives.
- ✓ Following the linear arrangement of the symbols, they are grouped by boxes. Whereas the consensus agrees that each box contains the syllables that would make up a word, we state that each box contains the symbols that make up a complete **sentence**.
- ✓ We have seen that there are conceptual **phrases** that are captured in a single sentence, whereas others need two or three sentences to be deployed.
- ✓ It is observed that, from time to time, a diagonal or vertical mark appears over the first symbol of a sentence. They were clearly made after the stamp marking. Once making the translations and putting the texts together, it became evident (clearer on side A) that they are a **mark of "new paragraph"**. This means that sentences are organized into **paragraphs**, and a sentence marked in this way is the last sentence of the current paragraph. These marks had been noticed before[7] and the hypothesis that a kind of paragraphs might exist was launched. In this moment, as we will see, we can doubtlessly confirm this point.
- ✓ Finally, just before the first symbol of the first sentence of side A, "flower", there seems to be a dot mark, which could well indicate the **beginning** of the text. In this case, and we might be wrong, we think this dot mark hadn't been noticed before.

PRELIMINARY CONSIDERATION

The fact that phrases and paragraphs appear, and that some of the phrases and words are repeated on a regular basis, makes the original message appear to have a certain rhythm or

[6] The spiral is a typical growth pattern in nature: from the inside out. And e.g. the arrangement of symbols in sentence A3 would already be obvious proof, because it shows that they ran out of space at the end of this sentence.

[7] Arthur J. Evans, Duhoux, Ohlenroth and others, "Scripta Minoa, the written documents of Minoan Crete, with special reference to the archives of Knossos" (1909). Oxford Clarendon Press. OL 7128491M.

cadence. As does any current prayer or prayer of the religious confession that we have been taught as children.

Here, obviously, when interpreting the ideograms, and putting these ideas together to form sentences in our own language, the original language is completely lost. The way this people had to refer to these concrete concepts. The idioms that their oral tradition had brought to them.

Today, and using an ideographic approach, the reproduction of such an original language certainly seems impossible. We can only confirm that the intention of the disc is to support a specific message that we can now understand, at least conceptually.

PROPOSED MEANING OF SYMBOLS

Below are the symbols[8] used on the two sides of the disc one by one according to their **order of appearance**, from the centre of the side A to the outer end of the side B.

Up to **46 different symbols are identified**[9], to which a number, identifying name, and a meaning is assigned.

There are pictograms that are very explicit and others that are not so explicit, normally ideograms, which usually represent pronouns, conjunctions, etc. In all possible cases, the reasoning followed to arrive at the proposed interpretation is explained.

The set of meanings shown here is the result of successive revisions to achieve the most coherent and appropriate meanings set considering the context of this artefact. At this point it is not intended to enter discussions about this or that meaning, as this would be a minor objective compared to the fact of checking that the disc can be actually read and understood.

#	Symbol	Name	Meaning
1		Flower	= **OH! / WE PRAY** An 8-petaled flower is shown. It is not an explicit symbol, but it has been derived from the meaning of the other symbols in the sentences. It refers to the manner of addressing or praying to the chief deity.
2		Goddess	= **GODDESS** It depicts an elaborate human head without hair, which would indicate purity[10], and with what appear to be earrings. Being the one to whom the prayer is addressed from the very beginning, she could be the chief deity. It is not the aim of this study to find out which goddess it is referring to.

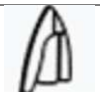
[8] For this study, drawings of the disc symbols that have been taken directly from the Wikipedia website (https://en.wikipedia.org/wiki/Phaistos_Disc) are used as a reference. The drawings correspond to the signs that were added to the Unicode (Universal Computer Character, UCS) character set in 2008. As the current consensus was that text should be read clockwise, from the outside toward the center, it looks that it was decided that these symbols in Unicode should be flipped relative to their orientation on the disc. Nevertheless, for this study, symbols are presented just as they appear on the disc, also coherent with the counterclockwise reading that we firmly suggest.

[9] All available documentation talk about only 45 different symbols. We bring to light a new one, identified in the above list as No. 45, to be different to the No. 39, mainly due to a different size of stamp and some image alterations.

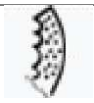
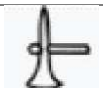
[10] Lots of explicit references here, e.g. in “Isis and Osiris” book from Plutarch’s Moralia.

3		Oar	= **TO RULE / TO GOVERN** (action) The oar or rudder steers a ship. Since Crete is an island, it seems like a very explicit image: whoever holds the oar or rudder has the power over the others, rules.
4		Person	= **PEOPLE** Pictogram representing in a generic way, not individually, both men and women, children, and the elderly.
5		Sceptre	= **ALL** It's not a very clear image, but it's derived from the meaning of the rest of the symbols in the sentences. It could likely represent the sceptre that kings carried in their hands and that meant holding power over all the people. It is always shown next to symbol “person”, i.e. ALL THE PEOPLE.
6		Bird	= **SKY / HEAVEN** A highly stylized image of a large bird flying high is presented (funny resemblance with Nazca lines). It is identified with the idea of heaven as a generic place.
7		Tree	= **EARTH** We see the schematic representation of a tree. It is identified with the idea of the earth as a generic place. It is depicted next to previous symbol, i.e. HEAVEN & EARTH.
8		Leaves	= **TO SPROUT / TO SHINE / TO PROVIDE** (action) Leaf buds are seen on a young plant. It is identified with the action of sprouting, arising, providing something.
9		Hide	= **WEALTH / RICHES** The skin of an animal is shown. It is identified with the benefit that remains after performing an activity, it is related to the success of an activity, the idea of goods or wealth.
10		Shield	= **TO LOOK AFTER / TO PROTECT** (action) Pictogram of a shield. The shield protects its wearer. This symbol is always accompanied by another symbol that indicates who it protects.
11		Man	= **MEN / US** It is understood as a generic, non-individual concept, referring to the group of men, as the main authors and possible beneficiaries of the prayer to the gods. It usually appears after "shield", which would stand for PROTECT US.
12		(Y-shaped) Stick	= **YOUR (HOLY) WILL** It's a symbol that's not explicit at all (at least in our culture). Its meaning is derived from the rest of the symbols.
13		Mace	= **WITH STRENGHT / A LOT / MANY** A mace or hammer is presented, which would indicate a forceful action, or a forceful result.
14		Snake	= **ILLNESSES / EVIL** The crawling serpent would generically represent all those evils that lurk.
15		Hawk	= **FIGHT / DELIVER US FROM** (action) A bird of prey is depicted carrying some prey in its talons. It can see snakes from above and hunt them. This symbol is always used after the "snake", so they should be understood together, i.e. FIGHT AGAINST EVIL / DELIVER US FROM EVIL.

16		Seed	= **FERTILE / FERTILITY** An ideogram with not obvious meaning. It can be derived from adjacent symbols though. It may be a representation of a bulge in a belly caused by a growing human fetus.
17		Arrow	= **ALSO / FURTHERMORE** This ideogram is not explicit and has been easily derived from the meaning of the other symbols in the sentences. It represents a concatenation of ideas, in relation to the previous sentences.
18		Woman	= **WOMEN** It's the most sophisticated image. Although it is not well seen in this drawing, the woman seems to be sitting, and on her lap she would have perhaps a small animal that she caresses. This symbol is used to differentiate from the symbol "man" when referring to fertility. There is also an occasion when symbols men and women appear together.
19		Pigeon	= **TO RAISE / TO GROW** (action) The pigeon or bird can take flight, soar, and is almost impossible to catch and stop. It would be an ideogram of the action of raising, making something grow.
20		Glasses	= **TO MULTIPLY** (action) This ideogram appears to show something that appears duplicated. Generically, it would be the action of multiplying.
21		Altar	= **TO WORSHIP / TO PRAY** (action) This symbol seems to represent an altar (e.g. the Altar of Malia, Crete) with an idol on top. It always appears after the symbol "person", so it should be understood as a whole and would mean PEOPLE PRAY TO YOU / PEOPLE WORSHIP YOU.
22		Ship	= **MERCHANT FLEET / DESTINY** A ship with its rudder is seen. It could refer very explicitly to the maritime expeditions that these people organized (the truth is that this explicit meaning works very well in a possible translation); or more likely, it symbolically refers to the idea of destiny, after the idea that goddess is at the helm.
23		Fish	= **(FRESH) FOOD** Pictogram with explicit representation of a fish, although it is understood to refer generically to fresh food.
24		Bow	= **STORAGE** Although this drawing is not very clear, the disc clearly depicts a loose arch/bow. It is an ideogram interpreted as the possibility of reaching something in the distance, going far.
25		Grain	= **GRAIN** This explicit image of a plant suggests something related to plantations or food. But it differs from the symbol "fish", (fresh) food, so it can be the grain that can accumulate and is consumed over time. Next to "bow" could refer to GRAIN RESERVES.
26		Bell	= **TO ASK FOR / TO WANT** (action) The symbol looks like a bell with which e.g. to alert or call the people to an event. It would indicate a request to the deity. It is like the "flower" symbol, but less important.

27		Double	**= OUR** The ideogram is not very explicit. The meaning is simply derived from the rest of the phrase symbols where it appears. It can be understood to represent something that encompasses all of yours, along with all of mine; our thing.
28		Logs	= **LOVED ONES** The drawing is not very clear, but you can see on the disc two vertical trunks completely joined. It is an ideogram interpreted to mean a very close personal relationship.
29		Crown	= **KINGS / LEADERS** The symbol is interpreted as a crown of feathers that would indicate a generic reference to a main character, such as the leaders of the different domains or kingdoms.
30		Head	= **FOREIGNER / NEIGHBOUR / OTHER** The symbol seems to represent a very unelaborated human head, which could indicate the idea of a foreigner/neighbour being, or perhaps just the idea of a generic other.
31		Water	= **SEA / DEITY OF SEA** Being a people dependent on the sea, one of their main deities would be associated with the sea. It is referenced from the end of Side A and all over Side B. Again, it is not the goal of this study to find out which deity this is referring to.
32		Prisoner	= **ENEMIES** [11] It appears to depict a man with his hands tied, i.e. a prisoner. It would refer generically to all enemies.
33		Beetle	= **TO SEARCH / TO FIND** (action) The image of the beetle, permanently moving, is used to suggest the action of looking for and/or finding something.
34		Dog	= **FRIENDS / ALLIES** A pictogram of a dog's head is used very explicitly to represent the idea of friendship. It is also used either as a noun or as an adjective to describe other symbols.
35		Palace	= **DOMAIN / KINGDOM** It is a very explicit symbol of an important house, perhaps a palace, which would make generic reference to the entire community, town, or kingdom.
36		Goat	= **HERD / ANIMALS** It is the head of an animal, perhaps a goat, which may explicitly refer to the concept of herd.
37		Hood	= **TO INTERCHANGE / TO TRADE** (action) The meaning of this symbol was derived from the context of the phrases. It is interpreted to correspond to the action of exchanging or trading goods. The pictogram could refer to the act of exchanging one hat for another.
38		Branches	= **TOGETHER / COMMON** Here are two branches of a tree growing from a common trunk. It is understood as a relationship established between two parties with a common interest.

[11] This symbol appears badly damaged at the outer end of the side A. One might think that it is the same No. 46 "kid" symbol with added marks on it, but it turns out to be a different size, so it is a different stamp and therefore a different symbol (see Appendix A).

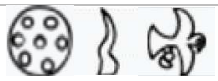
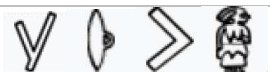
39		2-Legs	= **FUTURE / PROSPERITY** The meaning of this symbol was derived from the context of the phrases. This image appears to show a pillar or support with its two legs. Or it could be the representation of a human trunk with 2 legs. It could refer to the prosperity / future of the people that would be sustained by firm foundations or perhaps, that is done walking forward.
40		Scythe	= **HARVEST / TO COLLECT** (action) A sickle-type tool is explicitly presented, perhaps for harvesting, so it would indicate the concept of gathering or the action of harvesting the crop.
41		Shell	= **NEW** A seashell is explicitly presented. Conch shells are discovered and found on the shores, so it may represent the concept of something new.
42		Fruit	= **ORIGIN / THE BEGINING** The meaning of this symbol was derived from the context of the phrase in which it appears. In this drawing it is not well appreciated, but the disc shows a fruit or tuber that has germinated, from which several shoots come out. Ideogram that could suggest the idea of the beginning, origin, or germ.
43		Axe	= **PEACE** This explicit symbol would be an axe, which may very well be a weapon of war. But the fact that it is upside down could indicate that the concept referred to is that of peace.
44		Triangle	= **VEGETABLE PATCH** The meaning of this symbol was derived from the context of the phrase in which it appears. The pictogram would show a cultivated field. Also, the shape of an inverted triangle could also have to do with the complementary idea of fertility[12].
45		3-Legs	= **FAVOURABLE / SUCCESSFUL** It is not very explicit and appears in a very short sentence, which makes it difficult to interpret. It is derived from "2-Legs", then they must be related[13]. The symbol appears to show a third leg, which is understood to represent an advantage. The interpretation we give it is that it can refer to an action being lucky or successful.
46		Kid	= **CHILDREN** This explicit pictogram shows a boy, although it is understood that it refers generically to children.

READING THE SIDE "A" OF THE DISC

Below we are going to read one by one the **31 sentences** that appear on the side A of the disc.

[12] See the final Conclusion chapter for more information on the possible origin of this symbol.

[13] One might think that it is the same symbol as No. 39, "2-legs", but it has a different size and the drawing has some differences, so it is a different stamp and therefore a different symbol (see discussion in Appendix A). See also more information related to the historical significance of this new symbol in Appendix B.

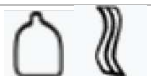
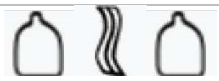
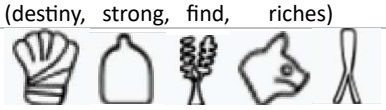
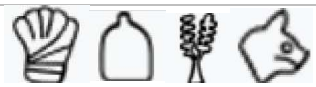
#	Sentence	Possible free translation
A1 .	(we pray, Goddess, govern)	We pray (Oh! Mother) Goddess, you who govern-
A2	(people, all)	-Over all the people.
A3	(heaven, earth, sprout, riches, riches, protect, men)	May the heaven and earth provide us with their riches, (mother) protect us.
A4	(we pray, Goddess, govern)	We pray Oh! Goddess, you who govern,
A5	(sprout, your will, strong)	-May your strong will-
A6	(people, all, protect, men)	-Over all the people, (mother) protect us.
A7	 (protect, evil, fight)	Help us to overcome evil.
A8	 (your will, fertility, also, women)	May your fertility will also be reflected in women.
A9	(riches, also, grow, multiply, riches)	May our riches also grow and multiply.
A10 \	(evil, fight, protect, men)	And overcome evil, (mother) protect us.
A11 \	(people, pray)	Our people pray to you.
A12	(also, strong, govern, destiny, riches, men)	Furthermore, may your strength guide our destiny to riches.
A13 \	(evil, fight, protect, men)	And overcome evil, (mother) protect us.
A14	(strong, food)	May we have a lot of food.
A15	(heaven, earth, sprout, riches, riches, protect, men)	May the heaven and earth provide us with their riches, (mother) protect us.
A16 \	(evil, fight, protect, men)	And overcome evil, (mother) protect us.

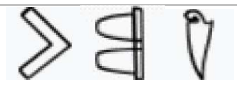
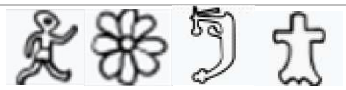
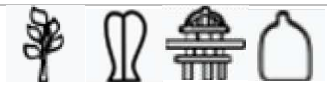
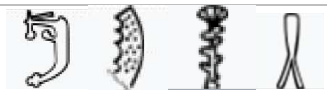
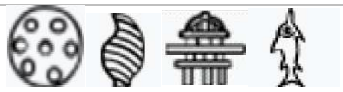
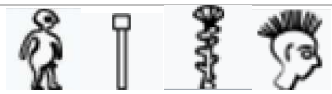
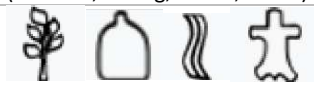
A17 \	(people, pray)	Our people pray to you.
A18	(also, strong, govern, destiny, riches, men)	Furthermore, may your strength guide our destiny to riches.
A19	(storage, grain)	May the grain storage-
A20 \	(we pray, strong, grow, protect, men)	-we pray, grow strong, (mother) protect us.
A21	(ask for, our, loved, people)	We ask that our loved ones-
A22	(sprout, your will, loved, protect, men)	-Be provided with your loving will, (mother) protect us.
A23	(sprout, evil, fight)	And deliver us from evil.
A24 \	.... (....., ..., also, women, protect, men)	[14] May your fertility will be also reflected in women, (mother) protect us.
A25 \	(kings, foreigner, riches)	About the foreign leaders' riches.
A26	(protect, ask for, Deity of Sea, riches)	(Oh! Mother) Protect, we ask you Deity of Sea, their riches-
A27	(food, our, enemies, protect, men)	-and their food from our enemies, (mother) protect us.
A28	(find, friend, friend)	May we find more friends.
A29 \	(ask for, Deity of Sea, friend)	We ask Deity of Sea for the friend.
A30	(protect, our, kingdom)	(Oh! Mother) protect our kingdom-
A31 \	(also, people, all, protect, men)	-And also all the people, (mother) protect us.

[14] Here the disc is badly damaged. We assume that e.g. it is a repetition of A8, and that it bears the diagonal mark.

READING THE SIDE "B" OF THE DISC

Below we are going to read one by one the **30 sentences** that appear on the side B of the disc.

#	Sentence	Possible free translation
B1 \	 (ask for, Deity of Sea)	We ask you Deity of Sea.
B2	(destiny, strong, find, friend)	The destiny may bring (us) many friends.
B3	(ask for, strong, sprout, women, men)	We ask you that lots of women and men may turn up.
B4	(ask for, also, grain, herd, trade)	We ask they also want to trade with grain and animals.
B5 \	(kings, ask for, joint, friend, prosperity)	Their kings may want to share prosperity as allies.
B6	(kingdom, also, strong, ask for)	So their kingdoms are also strong, we ask.
B7 \	 (ask for, Deity of Sea, ask for)	We ask Deity of Sea, all we are asking-
B8	(sprout, also, ask for)	-May become true, and we also ask-
B9	(destiny, strong, find, riches)	-The destiny may provide them with many riches.
B10 \	 (kings, ask for, joint, friend, prosperity)	Their kings may want to share prosperity as allies.
B11 \	(ask for, Deity of Sea, friend)	We ask Deity of Sea for the friend.
B12	(all, kings, friend)	For all the leaders are friends.
B13 \	 (kings, ask for, joint, friend)	Kings may want a common friendship.
B14	(people, riches, trade, men)	People may trade riches with us.

B15	(food, grain, grow, sprout, women)	So the food and grain may grow and provide to women.
B16	(people, food, friend)	May all our friends be provided with food-
B17	 (also, multiply, harvest)	-And also may the harvest be multiplied.
B18	(sprout, new, kingdom, kingdom, friend)	That new friendly kingdoms emerge.
B19	 (people, we pray, destiny, riches)	Our people pray (oh! Mother), the destiny may provide riches-
B20	(our, joint, evil, men)	-And men may fight our common evil.
B21	 (sprout, our, kingdom, ask for)	Thus, our kingdom may flourish, we ask.
B22	 (destiny, origin, earth, prosperity)	May the destiny bring the origin of this land's prosperity.
B23 \	(also, people, all, ask for, peace)	And also everyone wants peace.
B24	(food, grain, people, all)	May there be food and grain for everyone.
B25 \	(cultivated patch, also, strong, harvest)	May the land also produce an abundant harvest.
B26	 (protect, new, kingdom, food)	(Oh! Mother) protect the new kingdom may it have food, -
B27	(riches, destiny, successful)	-Riches, and a successful destiny.
B28 \	 (children, strong, earth, men)	May the land's men have lots of children.
B29	 (sprout, ask for, Deity of Sea, riches)	That all these become true, we ask you Deity of Sea for riches.
B30	(ask for, our, prosperity, protect, men)	We ask you for our prosperity, (mother) protect us.

PRAYER (CONCEPTUAL) FROM THE PHAISTOS DISC

.	A1	We pray (Oh! Mother) Goddess, you who govern
	A2	over all the people.
	A3	May the heaven and earth provide us with their riches, (mother) protect us.
	A4	We pray Oh! Goddess, you who govern,
	A5	may your strong will
	A6	over all the people, (mother) protect us.
	A7	Help us to overcome evil.
	A8	May your fertility will also be reflected in women.
	A9	May our riches also grow and multiply.
\	A10	And overcome evil, (mother) protect us.
\	A11	Our people pray to you.
	A12	Furthermore, may your strength guide our destiny to riches.
\	A13	And overcome evil, (mother) protect us.
	A14	May we have a lot of food.
	A15	May the heaven and earth provide us with their riches, (mother) protect us.
\	A16	And overcome evil, (mother) protect us.
\	A17	Our people pray to you.
	A18	Furthermore, may your strength guide our destiny to riches.
	A19	May the grain storage,
\	A20	we pray, grow strong, (mother) protect us.
	A21	We ask that our loved ones
	A22	be provided with your loving will, (mother) protect us.
	A23	And deliver us from evil.
\	A24	May your fertility will be also reflected in women, (mother) protect us.
\	A25	About the foreign leaders' riches.
	A26	(Oh! Mother) Protect, we ask you Deity of Sea, their riches-
	A27	-and their food from our enemies, (mother) protect us.
	A28	May we find more friends.
\	A29	We ask Deity of Sea for the friend.
	A30	(Oh! Mother) protect our kingdom
\	A31	and also all the people, (mother) protect us.

\	B1	We ask you Deity of Sea.
	B2	The destiny may bring (us) many friends.
	B3	We ask you that lots of women and men may turn up.
	B4	We ask they also want to trade with grain and animals.
\	B5	Their kings may want to share prosperity as allies.
	B6	So their kingdoms are also strong, we ask.
\	B7	We ask Deity of Sea, all we are asking
	B8	may become true, and we also ask
	B9	the destiny may provide them with many riches.
\	B10	Their kings may want to share prosperity as allies.
\	B11	We ask Deity of Sea for the friend.
	B12	For all the leaders are friends.
\	B13	Kings may want a common friendship.
	B14	People may trade riches with us.
	B15	So the food and grain may grow and provide to women.
	B16	May all our friends be provided with food
	B17	and also may the harvest be multiplied.
	B18	That new friendly kingdoms emerge.
	B19	Our people pray (oh! Mother), the destiny may provide riches-
	B20	-And men may fight our common evil.
	B21	Thus, our kingdom may flourish, we ask.
	B22	May the destiny bring the origin of this land's prosperity.
\	B23	And also everyone wants peace.
	B24	May there be food and grain for everyone.
\	B25	May the land also produce an abundant harvest.
	B26	(Oh! Mother) protect the new kingdom may it have food, -
	B27	Riches, and a successful destiny.
\	B28	May the land's men have lots of children.
	B29	That all these become true, we ask you Deity of Sea for riches.
	B30	We ask you for our prosperity, (mother) protect us.

COMMENT

In view of the work presented, it will always be possible to discuss the proposed translation of any of the sentences. Or propose other alternative meanings for the symbols, which would alter the conceptual meaning of some of the phrases. But at this point, there is no doubt that **the ideographic approach works. And that the coherent and logical set of meanings that has been presented, together with some simple formal rules, yields a spectacular result.**

Despite the almost 4,000 years that have passed, it is impossible not to see a relationship between the current hymns and prayers with this one presented to us here on the Phaistos Disc. There are common terms that are very obvious to recognize in the text. As when it speaks of "rule above all people", "in heaven and earth", "to grow and multiply", "thy will be done", "deliver us from evil", etc. This strongly supports the idea that **the same formulas and traditions, first oral and then written, would have passed from civilization to civilization throughout history, thus being adopted by the different religions.**

Considering the form and structure of the text, it could be said that the first part is better constructed, following a clearer pattern. This could indicate that it is an ancient composition and that it would have been refined over time. However, the second part has a different, less elaborate structure. Which could also mean that it is a more modern composition than the previous one, and that it was added next.

Considering the meaning of the message, especially from the side B, we dare to launch a **hypothesis** that would place the creation of this disc within a very specific historical context. It could be a prayer made shortly after the island suffered the terrible catastrophic event derived from the eruption of the volcano on the neighbouring island of Santorini[15]. The key ideas that give rise to thinking in this sense are when it speaks of protecting the "new" kingdom (B26), that the kingdom "flourishes" or perhaps "resurfaces" (B21) and that destiny marks the "beginning" of prosperity (B22). This, together with the rest of the phrases dedicated to asking in general for new kingdoms, new people, that they be friendly, and have goods to trade and prosper in common etc., suggests that it could be referring to a situation in which the people are trying to recover and re-emerge after a catastrophe.

ABOUT CONSENSUS

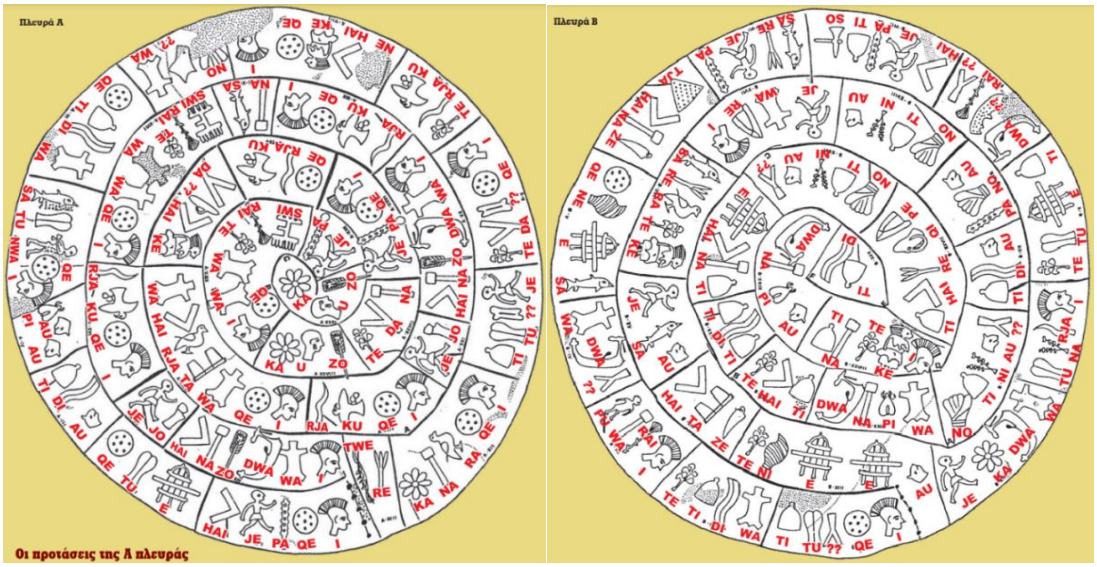
Now, in order to assess this result and compare it with the current state of the issue, we have to go to the website to look for the latest published information available. Among many others who have studied this topic for decades, the most notorious are currently three researchers: Gareth Owens (Associate Director, Hellenic Mediterranean University, former TEI Crete), John Coleman (Professor of Phonetics, Oxford University) and Christophoros Charalambakis (Professor of Linguistics, Athens University), who have been trying to decrypt the disc together since 2008. The current consensus is that each symbol should be a syllable or phonetic element. To do this, they presuppose that the Minoan script, which appears on the Phaistos Disc, on the Axe of Arkalochori, and the Altar of Malia, is somehow related to the Mycenaean Linear B syllabic script.

These researchers justify that the symbols must be syllables or phonemes mainly in the fact that, according to them, the 45 different symbols they have identified would be too many to constitute

[15] Dated to around 1,610 B.C. with a margin of error of +/- 14 years according to the Smithsonian's Global Volcanism program (https://es.wikipedia.org/wiki/Caldera_Santorini).

an alphabet, and too few to build a complete ideographic base such as the Chinese script.

They then opted for the syllabic approach and, based on "*epigraphic continuity and partial parallelism of some Arkalochori texts*"[16], assigned to each of the symbols on the disc (except 5 or 6) a syllable or phoneme. Thus, they claimed that the disc could be read, yet not understood at all, in about 80% of its length. In May 2014 Gareth Owens presented these results in a famous TED talk[17], generating numerous press releases saying that the disc had finally been decrypted.



18

According to these researchers, the symbols/syllables within a box would form a word, so the disc would contain 31 words on side A and another 30 on side B, following an orientation from the outside in. That TED talk ended with a metallic voice reading the syllables, pausing each supposed word, as shown in the photo above. We can only agree with Gareth Owens when, at his TED talk, he humorously said that with this translation they did not expect to win any Nobel Prize in Literature.

In order to appreciate the notable differences between the two approaches, a comparison of the first two words they would read is offered, starting with the end of side A.

A30				A31 \					
(protect, QE	our, TU	kingdom) E			(also, HAI	people, JE	all, PA	protect, QE	men) I

The current consensus indicates that the syllables/phonemes of these first two boxes of the disc, (inward orientation, from right to left), would form the following words supposedly in a language that should be very similar to Linear B, however without any meaning:

"/I-QE-PA-JE-HAI/ /E-TU-QE/..."

In contrast to the conceptual translation that we make of the ideograms:

"... (Oh! Mother) Protect our kingdom and also all people, (mother) protect us."

[16] View of the South magazine, Sep. 6th 2011 (https://daidalika.hmu.gr/wp-content/uploads/2020/11/mother.pdf).

[17] Gareth Owens, TED Talks: TEDxHeraklion, 23 mayo 2014 (https://www.youtube.com/watch?v=6Chcplx3tZ8).

[18] Drawing of the two sides of the Phaistos Disc with the syllables assigned to each symbol, taken from the website (https://daidalika.hmu.gr/minoan_scripts-bibliography/), "To read the Phaistos Disk" pdf link, ©Gareth Owens.

We can analyse the differences from numerous angles. Perhaps the most obvious is the practical point of view. It seems highly unlikely that anyone would take the trouble to design and create a stamp as complex as the figure of the man with hair, or the seated woman, or the palace, with only the intention of representing a syllable such as /I/, /KE/, or /E/ respectively. The amount of work, space, and resources to produce a single word would be excessive, making communication impossible. The entire disc would show only 61 words, i.e. about 3 or 4 lines of text.

Related to this criterion are the diagonal marks (aka "oblique stroke") that appear at the beginning of some sentences (9 or 10 on side A, 9 on side B), which Arthur J. Evans and others already suggested could represent the presence of paragraphs. In the phonetic approach they merely might "denote the ends of collections of signs group", i.e. an unspecific suggestion of verses and rhymes [19]. In our interpretation, these strokes are just "new paragraph" marks.

On the other hand, the presumption that the Minoan script could be syllabic, as Mycenaean Linear B is, does not seem to be very well founded: the fact is that Linear B will appear on the island several centuries after any form of Cretan Hieroglyph, and was brought by the Mycenaeans, a later civilization that overcome the Minoans[20]. Even if we could ignore this last, and in the logical development of a language, it usually takes hundreds of years to go from hieroglyphs (ideograms) to syllables (phonemes) to end up with a complete set of letters (alphabet). Generally speaking, it could be reasonable (same area, same civilization) that an ancient script such as the Cretan Hieroglyph, which includes the Minoan script, evolved into a syllabic system such as Linear B, perhaps through an intermediate one such as Linear A. But again, the fact is that, to date, there is not enough evidence to conclude that Linear A is syllabic. Therefore, the previous Minoan script should be even far less suspicious of being syllabic.

Reaffirming this impression is the fact that it has been many decades since Linear B was deciphered by Michael Ventris in 1952, and the great development of computers has allowed all kinds of exhaustive statistical analysis of languages: frequency of appearance of signs, coherence between texts, correlations with other languages, etc. Unfortunately, despite these advances, only a few steps have been taken in Linear A and this proposal by Gareth Owens in 2014 on the Phaistos Disc. In addition, and more remarkably, since then there has been no other relevant advance or publication, after almost ten years that all this information was made public and accessible to researchers. This could strongly suggest that the syllabic approach might not be the correct one to deal with the Minoan script and perhaps Linear A as well.

Finally, in the scientific world, after the studies of Physics of this author, there is a commonly accepted rule that, in short, says that when it comes to solving a problem you have to avoid any gratuitous complexity unless simpler solutions do not work. Or in other words, if two theories give the same or similar results, the simplest solution is probably the right one. It is known as the "Ockham's razor" principle. In this case of the Minoan script and the two different ideographic and syllabic approaches, since not even the premise that both give comparable results is fulfilled, the complexity of the syllabic approach would not be justified. Especially after the ideographic approach produces a very beautiful result, totally coherent with the context, and with no possible comparison with the syllabic one.

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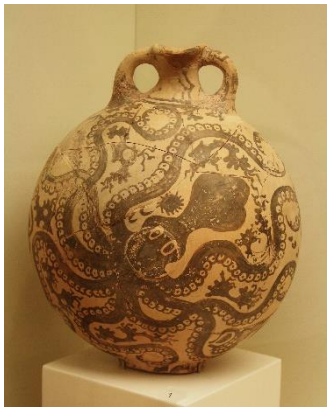
[19] Gareth Owens, "Evidence for the Minoan Language (5)-The Phaistos Disk. Chapter 6, To read the Phaistos Disk". Taken from web Daidalika. Link: (https://daidalika.hmu.gr/minoan_scripts-bibliography/Ph.D. 2 (1996)).

[20] We think that suggesting that Minoan script could be syllabic as Linear B is just because Mycenaeans brought it to Crete some centuries later, is a weak historical argument actually not very far away of saying e.g. that Mayan or Aztec script could be alphabetic as Spanish is, just because it was brought to the area by Cristóbal Colón in 1492 A.C.

ABOUT MINOAN POTTERY AT PHAISTOS



Like the decoration of the mural frescoes found in the various Minoan palaces, Minoan pottery generally presents natural decorative motifs, especially floral and, characteristically, from the sea (see photograph of an octopus)[21], presented in a free form and without showing any special symmetry, unlike later Mycenaean pottery.

But specifically in the archaeological site of the palace of Phaistos, a few pieces have been recovered where the use of seals or stamps similar to those used in the disc is evident. It would be a novelty never before seen in the history of ceramics. In this case, they would not have a communication use but purely decorative. In the photograph shown below, it can be seen that stamps have been used to replicate the same figure around a vessel. We see examples of the use of stamps similar to those already seen of the "flower", "goat", "dog", "triangle" and "woman".

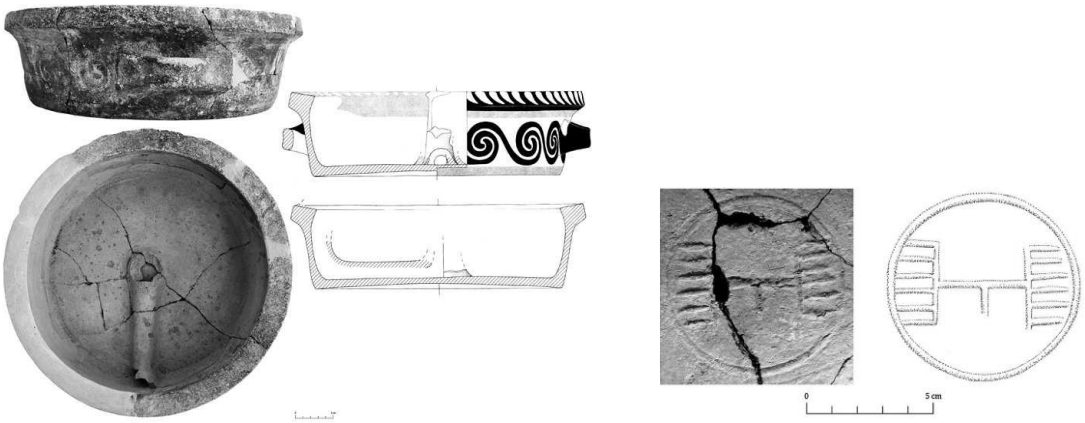
[22]

However, apart from a vessel in which a stamp very similar to that of the "shield" was used and which appeared in the palace of Knossos, no other examples of this type have been found, which would indicate that this artistic innovation did not have continuity.

[21] Picture taken from the web (https://www.worldhistory.org/image/367/minoan-vase-in-marine-style/). Uploaded by Mark Cartwright. License: Creative Commons Attribution-NonCommercial-ShareAlike.

[22] ©Giorgia Baldacci, "The Phaistos Disk-An Enigmatic Artifact and its Cultural Context". ASOR, The Ancient Near East (ANE) Today, Vol.9, No.11, November 2021: Comparisons between the Phaistos Disc signs and impressed ware motifs (image courtesy of A. Sanavia).

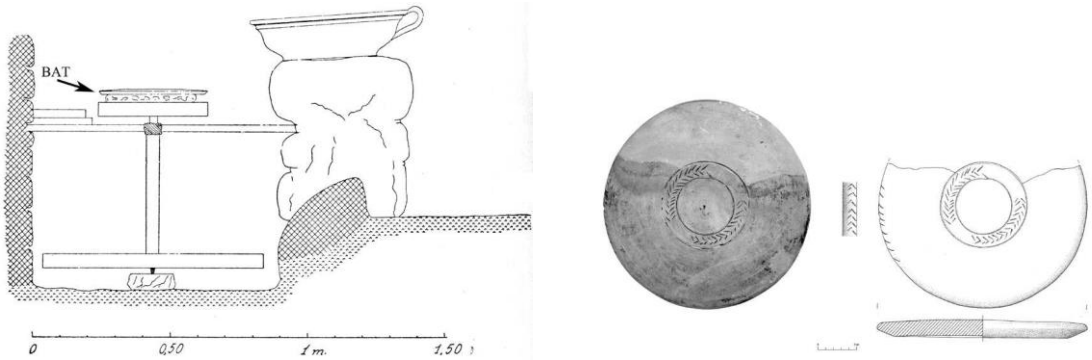
On the other hand, only a few examples of pottery have been found that show some of the Minoan symbols that we have seen on the disc. For example, the following photographs show a vessel, known as "Bowl F 4718" that bears the symbol of the "bird" marked underneath (like today's plates or vessels), in the typical Phaistos way. It seems that a stamp was also used here, although different from the one used in the disc. It could well be a kind of "made in Phaistos" mark, and perhaps it was carried by the ceramic utensils made in this palace. Or perhaps it was used occasionally as a mark that referenced or provided some magical aspect associated with the symbol. Unfortunately, no other objects with this same mark have been found.



23

A pottery turntable or bat (F 2298) has also been found, where the clay can be placed on top to model the different utensils, showing twice the symbol of the "tree" (earth), in a format of a single trunk or branch more similar to the one that can be seen e.g. on the Altar of Malia (see later) than the one shown on the Phaistos Disc. Here a seal or stamp was not used, but the symbols were marked by hand.

This object also seems to us to be a validation of the meaning, earth, that we gave to this sign at the time.

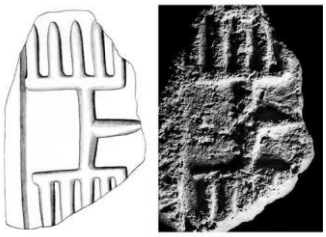


24

[23] ©Giorgia Baldacci, "Low-Relief Potters' marks and the Phaistos Disk: a note on the Comb sign". Annuario della Scuola Archeologica di Atene e delle Missioni Italiane in Oriente 95, 2017: Picture of Bowl F 4718 from Casa a Sud della Rampa, Phaistos (photo by the author; drawing by G. Merlatti).

[24] ©Giorgia Baldacci, "Low-Relief Potters' marks and the Phaistos Disk: a note on the Comb sign". Annuario della Scuola Archeologica di Atene e delle Missioni Italiane in Oriente 95, 2017: (left) Potter's wheel with bat, as observed in Cretan traditional workshops in the 1960s (after Fiandra - Pelagatti 1962; re-elaborated by the author). (right) Bat with incised motif from Phaistos (F 2298) (photo by the author; drawing by G. Merlatti).

Finally, we show the photograph of a small fragment of clay, known as "Sealing CMS II.5, n.246 (Phaistos)", showing the symbol of the "bird"[25] (heaven), in the typical form of Phaistos, where it seems that a stamp not identical but similar to that of the disc would have been used. It is believed that it may have been part of a larger object, probably a seal, and that it probably included more symbols. But since only this single symbol is now visible, the present object is not considered as another example of Minoan writing, as can be considered the fragment known as HM_992 that we will see later in this work.



[26]

Unlike the many hundreds of remains of pottery that have been found throughout the island of Crete with inscriptions from the Cretan Hieroglyph or Linear A, hardly any ceramic remains with these symbols of Minoan writing have been found.

This lack of pieces in the archaeological sites would show that this writing most likely never came into widespread use and its temporal validity was limited.

[25] It is very striking that since the first research works on the Minoan script, this symbol, which is one of the most characteristic of this writing, has always been known as "comb". It cannot be denied that it bears a certain resemblance to this object; at least one of the sides, because the image would be more like a double comb, which doesn't make much sense. It is also true that it is just a simple name of reference. But names have their importance, especially if they are applied to symbols that can be hieroglyphic in type, because names induce some kind of meaning. In this case, we do not believe that this name suggests or contributes anything in particular. Taking a look at the existing documentation, we find it very curious that since then no one has proposed or published the idea that this symbol could represent a "bird", as it came to mind very clearly the first time we saw this image. Thinking about this theme, the image was related to a bird probably because of the great resemblance to the figures of Nazca, Peru, which show different animals represented in a very schematic way. And recently we have realized that these figures were rediscovered in the decade of the 30s of the last century and had a progressive diffusion from the following decade of the 40s. Which is why they could not be present in the collective imagery of the first researchers at the beginning of the twentieth century. The very point here is that it seems that (all) the researchers who came after adopted a continuity approach and did not question this idea or perhaps did not consider it relevant, probably because the most widely used working hypothesis was still that writing was syllabic.

[26] ©Giorgia Baldacci, "Low-Relief Potters' marks and the Phaistos Disk: a note on the Comb sign". Yearbook of the Archaeological School of Athens and the Italian Missions in the East 95, 2017: Sealing CMS II.5, N. 246 (length 1.3 cm, height 0.7 cm), from Phaistos (image courtesy of the CMS Heidelberg).

CHAPTER II. AXE OF ARKALOCHORI

INTRODUCTION

Another of the objects found on the island of Crete that presents this Minoan script is the so-called Axe of Arkalochori, after the place where it was found (year 1934). It is also displayed in the Archaeological Museum of Heraklion, Crete. It is the metal piece of a double-bladed axe (*labrys*), a typical weapon of the Minoans that, in its central part, at least on one side (we cannot find information about the other side), presents three columns of symbols arranged like the sentences of the Phaistos Disc.

Being carved one by one directly into the metal, the symbols are necessarily much rougher than those found on the disc. However, once the work of identifying and assigning meaning to the symbols on the disc has been done, now we can try to identify the symbols that appear on the axe and thus read its message.

27

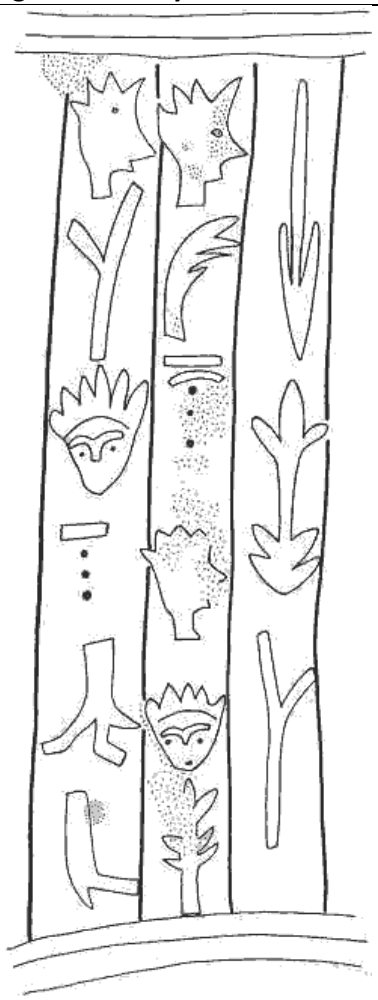
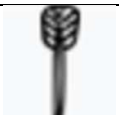
28

READING THE SYMBOLS OF THE AXE OF ARKALOCHORI

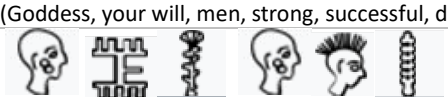
The axe has on one of its faces, in the central area, 15 symbols arranged in 3 vertical columns, framed within boxes as on the disc. Thus, we assume that they represent 3 sentences. The symbols should be read from top to bottom and from left to right.

Below is a table with the drawing of the symbols of the axe[29], and next to it a proposal of equivalence with the symbols and meanings that we have already seen on the disc.

[27] Axe of Arkalochori. Picture taken from Wikipedia with license Creative Commons CC0 1.0 Universal Public Domain Dedication. Link: (https://commons.wikimedia.org/wiki/File:Arkalochori_Axe_215.jpg).
[28] Axe detail. Link: (https://commons.wikimedia.org/wiki/File:Arkalochori_Axe_1c.jpg), same license.
[29] This table with the drawings of the symbols is used for reference only and has been taken from the Wikipedia website (https://commons.wikimedia.org/wiki/File:Arkalochori_Axe_(Inscription).png_/). Public domain: "faithful reproductions of two-dimensional public domain works of art are public domain" (Wikimedia Foundation).

Drawing of the axe symbols	Column 1	Column 2	Column 3
	(Goddess)	(Goddess)	 (govern)
	(your will)	(heaven)	(sprout)
	(men)	(earth)	(your will)
	(strong)	(Goddess)	
	(successful)	(men)	
	(destiny)	(all)	

Below is a possible free (conceptual) translation of the symbols.

#	Sentence	Possible free translation
1	(Goddess, your will, men, strong, successful, destiny)	It is the Goddess's will that we have a great successful destiny.
2	 (Goddess, heaven, earth, Goddess, men, all)	Oh! Goddess of heaven and earth, Goddess who above all men-
3	(govern, sprout, your will)	-rule, thy will be done.

MESSAGE (CONCEPTUAL) OF THE AXE OF ARKALOCHORI

1	It is the Goddess's will that we have a great successful destiny.
2	Oh! Goddess of heaven and earth, Goddess who above all men-
3	-rule, thy will be done.

COMMENT

The fact that we have clearly identified in the Axe of Arkalochori the formula already seen on the disc of "*goddess of heaven and earth, goddess who rules over all men, thy will be done*", which groups together a good number of symbols, helps us a lot to identify the differences in the presentation of the symbols of the axe with respect to those of the disc. For example:

- The axe represents the "Goddess" with a rather schematic head, with features that seem feminine to us, and with hair.
- The symbol of the "man" (us) is simpler and is represented as a forehead with a marked eyebrow, which would perhaps accentuate the idea of a male being. From the association of symbols that we have already seen on the disc, we would expect the symbol of "person" (people, generic). The change could be due to the fact that they considered alliances and wars to be the exclusive domain of men, not women.
- The symbol of Heaven is represented by a large bird's wing, simpler than the "bird" on the disc, it is supposed to simplify the execution on this material.
- The images of the "mace" (strong) and the "tree" (earth) are very schematic. This can be understood in the case of the figure of the tree due to its complexity, but not so much in the case of the mace. It indicates that on the disc there was a conceptual evolution of these symbols "mace" and "tree", since they seem to share part of the original design.
- The symbol that is identified as the "oar" (to govern) would be upside down with respect to the one we saw on the disc. It has not been possible to ascertain the fact that presenting a symbol up or down implies a change in its meaning. Except perhaps in the case of the "axe", which appears on the disc upside down (peace), which could have been used in some other Minoan text, not known, upwards and meant the opposite.
- The symbol "leaves" is somewhat different but is still perfectly recognizable.
- Outside of the aforementioned formula, we have the fourth symbol in the first column, which we identify as the "3-legs" sign of the disc. It could also be "2-legs", but the explicit figure on the axe showing the 3 "limbs", and how well the meaning fits into the context of the artefact leaves us in no doubt. In addition, we had already seen together the symbols of "3-legs" (successful) and "boat" (destiny) on the disc (B27).

The fact that this object presents less elaborate images for example for the "goddess", the "man", "the oar", the "tree" and the "mace" could indicate that the axe is something older than the disc, which was made before these four symbols underwent an evolution towards more explicit symbols such as those shown on the disc.

The text that emerges on the blade of the axe tells us that its wielder is protected by the Goddess, and that this weapon of war would be a mere instrument to enforce her sacred will, with which this person or his army hopes to be victorious. Since we assume that not all soldiers carried weapons like this, it can be inferred that the axe may have belonged to a military leader. It could also have been a votive offering to the Goddess, asking for protection in some battle against enemies. Or it could have been a combination of both: first it was used in a battle and then it could be deposited on an altar as a sign of gratitude.

The sentence carved on this very particular object is conceptually very beautiful. And it seems impossible to have obtained such a result by the simple fact of putting together this or that meaning at random. This reinforces the idea that **the study carried out, based on finding the most appropriate meanings for pictograms/ideograms, is the right way to face this challenge**. And the result obtained again seems spectacular to us.

ABOUT CONSENSUS

Also in this object, in order to assess the differences between the result that has just been presented using the ideographic approach versus the syllabic approach, we searched the web for the reading that Gareth Owens's team achieves using this other, more sophisticated method.

Unlike on the disc, here they read in the same orientation as us, that is, from top to bottom and from left to right. They have assigned a syllable to each symbol, except for two of them that they do not recognize and mark them with an /X/, and thus compose three supposed Linear B words without any meaning (see below screenshot taken from a video where they show the result of their work): "/I-DA-MA-NA-NI-[X]/ /I-NO-NA-I-MA-TE/ /[X]-ZO-DA/."

[30]

Looking at the syllables they read in this text and comparing them with those they assigned in their reading of the Phaistos Disc, it is easy to infer what has been the correspondence of symbols which they make between the objects. It is interesting to see the differences in interpretation.

Our proposal for the correspondence of signs with the disc:

Compared to the proposed correspondence from Gareth Owens's team:

Apart from being incomplete, there are numerous differences between their version and ours (in fact there are only 3 coincidences). The most relevant is in the first symbol that, while we clearly associate it with the Goddess (because of its more feminine than masculine features, because of the strong association with the next symbol, "stick" (holy will), and above all because of the long formula that is identified in the text), they associate it with the "man", instead of associating it with the male figure of the third symbol in the first column (this one does show clear masculine features). For this one, we see that it is translated as /MA/. But this syllable did not appear in the translation they made of the disc, so we cannot know which symbol they are referring to. It is intuited that they consider this masculine symbol of the axe as a new one to which, for convenience as we will see below, they assign this syllable /MA/.

[30] Screenshot of the video "Arkalochori Axe (2.014)", which appears as a link on the Daidalika website (https://daidalika.hmu.gr/home/) redirected to the YouTube link (https://www.youtube.com/watch?v=E72_fv1SYZs).

PARTIAL PARALLELISMS OF TEXTS IN ARKALOCHORI

The documents published by Gareth Owens's team suggest that their working hypothesis was that the Minoan script could be of the syllabic type due to the fact that it was a predecessor (although very distant) of the Mycenaean Linear B script, which is known to be syllabic. And in order to go back in time from Linear B, through Linear A, and finally arrive at the Minoan script, they said that they had relied on *"epigraphic continuity and partial parallelisms of texts in Arkalochori"*[31]. We found it interesting to look for information about this process.

Searching the web we first found a photograph of a golden axe ("AR Zf 1"), votive, also found at the archaeological site of Arkalochori, which shows four symbols originally attributed to the Linear A script, for which Gareth Owens's team proposes the supposed word "/I-DA-MA-TE/" (reading from left to right, again the opposite of what they said for the disc) as a translation. By following the syllable and symbol equivalences they made for the disc, it can be inferred which symbols they thought they were reading on this axe.

32

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However, we recognize here in this new axe four symbols that we have already seen on the disc, and so we can read and understand the message that is chiselled into it:

(Goddess, your will, find, earth)	Ideograms (free): "It is the Goddess's will that we find (new) lands/territories".
(¿?) / I - DA - MA* - TE /	Syllabic: "/I-DA-MA-TE/"

(*) Again, we can't identify the symbol they associate with /MA/. What it cannot be is the "beetle", since its associated syllable was /PI/.

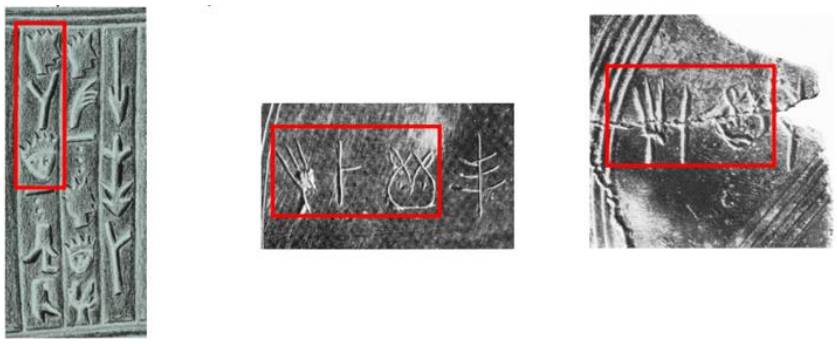
This new artefact seems very relevant to us because it implies that either these four symbols still belong to the Minoan script, or perhaps they are somewhat more recent and already belong to the beginning of Linear Minoan A. **But then it would show that this new script could be an evolution of the Minoan script and would maintain its ideographic characteristic**.

[31] View of the South magazine, Sep. 6th 2011 (https://daidalika.hmu.gr/wp-content/uploads/2020/11/mother.pdf).
[32] Golden labrys ("AR Zf 1") from Arkalochori, exhibited at the Museum of Fine Arts (MFA) in Boston. Link: (hhttps://mnamon.sns.it/index.php?page=Immagini&id=19&img=1216&lang=en).
[33] Detail of this golden axe, taken from the Daidalika website. Link: (https://daidalika.hmu.gr/minoan-linear-a/).

On the same website we later find a document that establishes a series of relationships between the symbols of the disc with others of the Cretan Hieroglyph and Linear A (some clearer, others very debatable). And these in turn would be connected with syllables of Linear B. Thus, they believe that they can relate Linear B first to Linear A, and then to the Phaistos Disc through the Axe of Arkalochori and the Altar of Malia that would serve as, they called, "epigraphic bridges"[34]. This is what they mean when they speak of a supposed **"*epigraphic continuity*".**

On the other hand, in this same document we see again the golden axe from before, along with a new third axe ("AR Zf 2"), this made of silver, not gold, that shows the same four symbols with some small variation. This document suggests that there would be an equivalence between the first three signs of these two new axes with the first three symbols of the (first) Axe of Arkalochori. When Gareth Owens's team speaks of "***partial parallelisms of texts in Arkalochori***" they must refer to this alleged equivalence, since there are no other possible texts.

The reference to "partial" parallels of texts in Arkalochori would come from the fact that they could not associate the fourth symbol of the two new axes (a more schematic version of the "tree" with a single trunk or branch, not the "leaves" as they identify) with the fourth symbol of the first axe (the "mace"), but only the first three. In the photograph shown below, taken from this same document mentioned above, the red boxes would indicate the supposed equivalence of the three symbols. If we analyse them:



35

- ✓ The first symbol that appears from the left on the two new axes is a more schematic representation of a human head. The question is to know which one, the feminine or the masculine.
    - If we follow the ideograms, because of the strong association with the second symbol, the head must be that of the goddess. Although it would be a more schematic version of the goddess as represented in profile on the (first) Axe of Arkalochori, not the one that appeared on the disc (hairless).
    - From the translation they offer, /I/, it follows that they think that this head is the same male head of the disc. In the absence of any semantic link, we suppose that they were guided more by the visual aspect, and it seemed to them that this new, more schematic figure reflected the head of the man with that characteristic crest that he presented on the disc. They apparently did not contemplate, or ended up discarding, the possibility that the first symbol of the (first) Axe of Arkalochori was actually the female figure of the goddess.

[34] Webpage Daidalika, Gareth Owens (1996), "Evidence for the Minoan Language (5)-The Phaistos Disk". Link: (https://daidalika.hmu.gr/minoan_scripts-bibliography/Ph.D. 1 (1996)).

[35] Photograph taken from the document "A table of similarities", found on the Daidalika website. Link: (https://daidalika.hmu.gr/wp-content/uploads/2020/11/signary_July2017b.pdf).

- ✓ The middle symbol is the one that causes the least problems, since there is no discrepancy in the identification, and it does seem to be the same in all three objects.
    - For us it is the symbol "stick" (holy will). On the disc it was almost always used together with the symbol of the goddess.
    - According to the syllabic interpretation, this sign has /DA/ syllable associated.

- ✓ The third symbol from the left on the two new axes is also problematic.
    - Because of its characteristic shape, we clearly recognize here the "beetle" sign (search/find) that we saw on the disc.
    - From the translation offered by Gareth Owens's team, /MA/, it is derived that for whatever reason, and despite there being no resemblance between them, they think that this symbol must be the same as the third one of the (first) Axe of Arkalochori. They interpret that it must be a new male figure that did not exist on the disc. It may have been convenient for them, but the truth is that it does not seem to be well justified.

When they proposed the translation "/I-DA-MA-TE/" for the 4 symbols that appeared on the new axes, what we wonder is where they got the syllabic element /MA/, since it was unknown until now. We assume then that this must be a word similar to one known in Linear B, or perhaps the name of some deity that they tried to identify here on this object[36]. And this may have had a great influence on how they later built their associations between symbols and syllables for the rest of the objects.

This is probably where the association of the first symbol of the axes with the male figure of the disc comes from. To which, among all the possibilities offered by an epigraphic study, they finally decided to assign the syllable /I/ (apparently, they hesitated between /A/ or /I/)[37]. Because this symbol "man" would be the beginning of many words on the disc (according to their interpretation and reading from right to left and from the outer edge to the centre) and they must have found it interesting to have this syllable /I/ as the beginning of all those words.

It also seems very possible to us that this idea therefore decided the orientation of the reading they made of the Phaistos Disc: from right to left. This orientation allows the appearance of the syllable /I/ as the first in the different supposed words of the text. Although, by the same rule, at least in the two new axes, the direction of the reading should be just the opposite, from left to right, now following the orientation of the human faces. But it seems that they did not give much importance to this possible formal incoherence.

In the end, however, in our opinion, these supposed equivalences or partial parallelisms of texts proposed by Gareth Owens's team would not be such. Two of the three symbols involved would be misidentified, so the conclusions that they were able to draw from this, and that they most likely applied in their study of the rest of the objects (in the Phaistos Disc and in the first Axe of Arkalochori), would unfortunately be wrong.

According to our ideographic interpretation, these three votive axes found in Arkalochori share the initial (conceptual) formula of "It is the Goddess's will that..." in their inscriptions. The first two symbols, the "Goddess" and "stick", would effectively be the same. This is the only partial parallelism that we observe in these three axes.

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[36] See a reference to "Demeter" explained on the website of Gareth Owens (https://daidalika.hmu.gr/demeter/).
[37] "Read Phaistos Disk?" (https://daidalika.hmu.gr/wp-content/uploads/2020/12/READNEW2020PD.pdf).

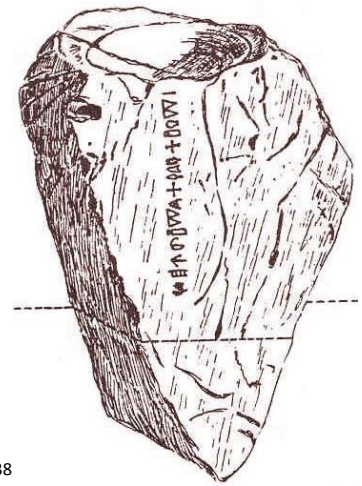
CHAPTER III. ALTAR OF MALIA

INTRODUCTION

Another of the objects found on the island of Crete that presents the **Minoan script** is the so-called Altar of Malia, after the place where it was found. It is a stone altar or pedestal, on the top of which a figure of a deity could be placed, or it could serve as a support for offerings. On its front face, symbols are carved arranged in a single vertical column.

This object was catalogued within the *Corpus Hieroglyphicarum Inscriptionum Cretae* (CHIC) as "CHIC 328 (Malia)".

Being carved directly into the stone, the symbols are necessarily much rougher than those found on the disc. However, once the work of identifying and assigning meaning to the symbols on the disc has been done, we can try to identify those that appear on the altar and read its message.



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READING THE SYMBOLS OF THE ALTAR OF MALIA

The stone ashlar has 1 column on its front face with 15 symbols in a vertical arrangement, unframed. Symbols should be read from top to bottom. It may be a simple coincidence, but it is the same number of symbols that appeared on the (first) Axe of Arkalochori.

Below is a table with the drawing of the 15 symbols of the altar[41], and next to it a proposal of equivalence with the symbols and meanings that we have seen on the disc.

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[38] This drawing that artistically represents the Altar of Malia has been taken from the website (https://mwenge.github.io/linear0.xyz/) programmed by Robert Hogan. All illustrations are copyrighted by the École française d'Athènes.

[39] Old picture of the Altar of Malia found in document "On sacred vocabulary and religious dedications: the Minoan libation formula" (ref: after MARINATOS, supra n. 73, 229-230, fig.8), Artemis Karnava (University of Crete), conference paper, April 2014. Link: (https://www.researchgate.net/publication/316285128).

[40] Detail the inscriptions on Altar of Malia taken from the website (https://mwenge.github.io/linear0.xyz/) programmed by Robert Hogan. All illustrations are copyrighted by the École française d'Athènes.

[41] The table with the drawings of the symbols that appear on the altar is used for reference only and has been taken from the website (https://mwenge.github.io/linear0.xyz/) programmed by Robert Hogan. All illustrations are copyrighted by the École française d'Athènes.

Drawing Symbols Altar	**Matching Disc Symbols**

- The symbol of the "bird" (heaven) changes again with respect to the disc and the axe. Here an ideogram of a four-pointed cross is depicted to indicate the sky.
- The "tree" (earth) symbol also changes a bit and is now represented by a single trunk or branch design.
- The "stick" symbol (your will) is the one that changes the most, as it is presented face down and as a closed figure. You might think that it resembles the symbol called "hood" (to trade), but it was dismissed both because of the lack of context in the phrase, and because of the frequency with which this entire group of symbols appear together in many of the phrases that we have seen in the rest of the objects.
- The symbol of the "sprout" is again something different because of the need to be more schematic and simpler, although it is still distinguishable without difficulty.
- Outside the formula, the symbol of the "2-legs" (prosperity) is also different. The ideogram presented here seems to show a kind of support or ladder, or perhaps rather a door leaf. In any case, it seems to us a conceptually very similar image to the disc (you have to go out the door to go in search of the future or prosperity). In addition, we have seen before the symbols "2-legs" (prosperity) and "branches" (common) together in the sentences B5 and B10 of the disc.

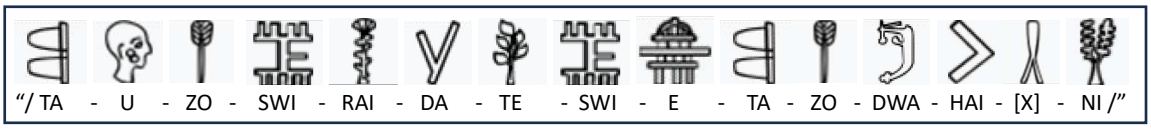
Considering the meaning of the phrase, the beginning of the phrase is striking, when it seems to summon all the deities. Until now, only references to a single chief goddess and to the deity of the sea had been seen. It is not possible to know if it refers to the fact that this people really had many deities, or if it is a matter of mentioning together all the gods or goddesses that each of the multiple kingdoms that are supposed to form an alliance had, and for which a common prosperous future is requested. The message that can be read on this altar is very coherent with the object and the context that we know of this people.

Again, arriving at this interpretation of the Minoan text does not seem likely to be the result of chance. **Especially if we already look at the 3 artefacts as a whole.**

ABOUT CONSENSUS

In this case we cannot make any comparison between our interpretation and the current consensus because **we have not been able to find, among the researchers who support the syllabic approach, a single attempt to read this inscription. Probably because it would be very difficult for them to explain that this object shows a single word with 15 syllables**.

Assuming that you wanted to read from left to right, and for example with the same equivalence of symbols of the disc and syllables that Gareth Owens's team did (or any other), the following supposed mega word would be formed in Linear B, obviously without any meaning:



This strongly suggests that what is actually being represented here by these Minoan symbols are ideograms, not syllables, and that they would form a sentence, not a word.

This object *per se* is further evidence that the syllabic approach does not seem to be the correct one to understand either this inscription or, by extension, the previous ones.

CHAPTER IV. FRAGMENT HM 992

INTRODUCTION

It is the last known object to feature a **Minoan inscription**. The one known as Fragment HM_992 is a piece of clay that was also found in the palace of Phaistos, the same site as the Phaistos Disc, although many years later, in 1955.

On one of its sides, it shows symbols similar to those on the disc. It is of a very small size, and it is clear that it was part of a larger piece, probably of a clay seal. What it doesn't seem like is that it's part of another disc similar to the Phaistos one.

[42]

READING THE SYMBOLS OF THE FRAGMENT HM_992

On the small piece of clay only three non-aligned symbols can be distinguished:

- Very clearly you can see a part of the symbol that we call "bird" (heaven) in the same format as the Phaistos Disc, thus showing what seems to be a local style.
- In the upper right you can also see what could be a large part of the symbol we call "tree" (earth). Not so much in the form of the Phaistos Disc, but perhaps more in the form presented on the Altar of Malia, that is, as a large tree with a single branch. It appears next to the heaven symbol, which is not surprising because the vast majority of the time these two symbols appear together (heaven and earth).
- And finally, just below the tree it shows part of a new symbol, which we have not seen so far in the three previous objects, in the shape of two concentric circles.

Being such a small piece, it is impossible to establish a context or a minimum meaning.

COMMENT

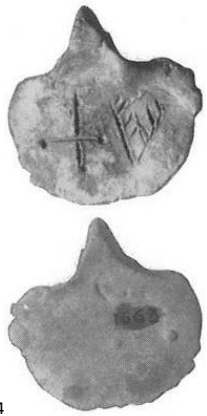
Attempts have been made to position this fragment on top of the Phaistos Disc, in case it was part of a copy of it, but no concordance is achieved. Partly because of the angle that the two main symbols present here to each other, and because there does not seem to be any conceptual relationship between any of the symbols adjacent to the heaven and earth appearances of the disc with this new symbol of the concentric circles.

Regarding the presentation of symbols, we see a curious issue that has just been mentioned above. The symbols of heaven and earth, which usually appear together and aligned in the sentences we have seen, in this case are marked on the clay in an odd way: if the "bird" were in

[42] This photograph of the fragment HM_992 has been taken from the website (https://ciphermysteries.com/wp-content/uploads/sites/6/2010/03/HM-992.jpg). The same photo is also on the page (https://the-phaistos-disk.webnode.page/).

a horizontal position, then the "tree" has been printed in a diagonal direction to the previous one, or the other way around: they are arranged at an angle of about 45 degrees between them.

But searching the internet has found a website that speaks of a collection of small objects that once formed what was called the Corpus of Cretan Hieroglyphic writing, published in 1909 by Arthur Evans (the year after the Phaistos Disc was found). In 1996 the last revision of this Corpus was published under the name of "*Corpus Hieroglyphicarum Inscriptionum Cretae*" (CHIC).[43] Among these objects or fragments, several of them are very striking, but for example and especially the one catalogued as "nº 81 (Malia)", within the so-called "group of clay fragments with inscriptions" [CHIC H 1-122].



44

This small object shows, on one side (the other is empty), an inscription of the **Cretan hieroglyphic script**. We may now suppose, without much discussion, that they correspond to the two symbols of heaven and earth which we have already seen many times together.

- ✓ The symbol of heaven in the form of a four-pointed cross as on the Altar of Malia.
- ✓ The earth symbol we see here represents a new variation on the "tree" symbol on the disc, and also on the one we see on the altar. But the curious thing is that the arrangement of the branches of the tree at an angle is reminiscent of what is seen in fragment HM_992.

Looking closely at the HM_992 fragment, the "tree" symbol might well originally have this shape seen in this CHIC_H_81 fragment, with the two branches. It is very possible that in this piece we are seeing only the branch on the left side and the one on the right has been lost. In fact, it seems to glimpse the beginning of this right branch. We have already seen that ideograms vary in shape from one object to another, following a kind of design evolution[45]. It would be reasonable e.g. for a first design of "tree" (earth) in two branches to then change to a single branch to avoid conflicts of interpretation with the other symbol called "branches" (common).

Although it is clear that the symbols in this fragment are not aligned, this could explain the odd angled arrangement of the two recognizable symbols in the HM_992.

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[43] Godart, Louis and Olivier, Jean-Pierre, Cretan Studies, 31, 1996, © 1996 - 2024 French School of Athens (https://cefael.efa.gr/detail.php?site_id=1&actionID=page&serie_id=EtCret&volume_number=31&issue_number=0&x=12&y=3&ce=smvhdocvggkceunck353728vv13tbthb&sp=443).

[44] Photograph of the object catalogued as "CHIC_H_81 (Malia)", front and back, taken from the website (https://mwenge.github.io/linear0.xyz/) programmed by Robert Hogan. All illustrations are copyrighted by the École française d'Athènes.

[45] Appendix D provides a table with the correspondence of symbols between all the objects we have seen in this study, and a possible evolution from perhaps the oldest to the most modern inscriptions.

CHAPTER V. CONCLUSION

THE MINOAN WRITING IS IDEOGRAPHIC

As has been defended throughout this study, the hypothesis that the Minoan script is of the syllabic type is, to say the least, very weak. This author understands that the main reason why a majority of researchers maintain this approach is that it provides them with a certain systematic scientific method that justifies their work in terms of better or worse ratio statistics: correlation between languages, appearance of symbols, frequency of use, habitual associations, etc. Which in the end does not prevent each reviewed work from including a good number of subjectivities.

This author recalls that, by definition, these ratio statistics are not infallible indicators of certainty and believes that it is a fundamental mistake to discard other approaches simply because they lack figures to support them. The difference is that consensus using these other approaches will surely be more complicated and will require considering a series of qualitative aspects other than mere statistical results.

Throughout this work several strong arguments have already been presented against of the Minoan script being syllabic: the beauty of our translation; the practical point of view of the resources; the new-paragraph marks; the fact that syllabic Mycenaean Linear B belonged to a later civilization; that nowadays there is no consensus that Linear A is syllabic; that since 1952 it has not been possible to decipher using this approach despite having all the computer support; by the criterion of "Ockham's razor"; the error of Arkalochori partial texts; and finally the example of the Altar of Malia where a supposed word with 15 syllables would be shown.

We now present here a new argument in favour of ideograms which seems to us definitive, and which we have recently encountered. Below are some photographs of some of the many "votive" figurines, all very similar, that appeared in 1971 at the archaeological site of Altyn-Depe, in present-day Turkmenistan, dated to the Middle Bronze Age (2,350-1,900 B.C.).[46]

47

48

[46] Lyubov B. Kircho, "Altyn-Depe, The Formation of the Earliest Urban Centre in central Asia", ©Harrassowitz Verlag, Wiesbaden 2019. Link: (https://www.researchgate.net/publication/337319374_Altyn-Depe_The_Formation_of_the_Earliest_Urban_Centre_in_Central_Asia).

[47] Terracotta figure from Altyn-Depe, exhibited at the ©Hermitage Museum, St. Petersburg. Photograph taken from their website (https://www.hermitagemuseum.org/wps/portal/hermitage/digital-collection/25.+Archaeological+Artifacts/1423930/?lng=en).

[48] More terracotta figures from Altyn-Depe, apparently sold online (493649, 494083 and 494086), e.g. link: (https://www.plazzart.com/en_NL/buy/archaeology/altyn-tepe-turkmenistan-fertility-goddess-ca-2000-bc-493649).

The unique thing about the case is that in all these figurines at least two of the symbols that we have already seen in the Minoan script are shown very clearly (marked by hand):

- the symbol "tree" (earth), in the very centre of the figure, in the format of a single trunk or branch, as it appears on the Altar of Malia or on the two axes of Arkalochori that supposedly showed Linear A symbols.
- and the symbol "triangle" (cultivated plot), as it appeared on the Phaistos Disc, unmistakable with that peculiar shape of the inverted triangle.

The figure leaves no room for doubt. It is very simple, but at the same time conceptually very beautiful. It is an image of the concept "Earth", in the form of a mother goddess to whom popular devotion was perhaps professed (due to the large number of figurines found), so that no one is confused it bears her name written on the chest, and presents the symbol of cultivated land: thus making explicit reference to the association of the idea of female fertility with the production of agricultural crops. The interpretation of this figure is very evident: they worshipped this goddess "Earth" to ask for her "favours" or "blessings" in the form of abundant harvests. One can only speculate about the extent of the fertility idea to which it refers was restricted only to crops or whether it implied a more or less broad concept also including perhaps livestock, people themselves, even rainy seasons, etc.

To our surprise, while we were looking for more information about these figures, we found a website where another new figure very similar to the previous ones was shown, also dated around 2,000 BC, but this time found in Charsadda Shaikhan Dheri, Gandhara, Pakistan, by an English soldier in the middle of the last century. The similarity is evident and the symbols it presents are very similar. Instead of having a support base, it has two perforations on each side of the head which indicates that it could go together with a rope or cord, perhaps as a pendant (a design that has also been seen in Altyn-Depe, see photographs). Apart from the simpler design, the execution is also somewhat rougher, which could suggest that it is older than the previous ones.

[49]

Apart from the exciting question that immediately arises as to how it is possible that in places that are in principle as far away as the island of Crete, Turkmenistan or Pakistan, more or less contemporary artefacts with the same symbols inscribed appear, and which is not the object of

[49] Terracotta figurine from Gandhara, on display at the © Victoria & Albert Museum, London. Photograph taken from their website (https://collections.vam.ac.uk/item/O120565/figure-figure-unknown/).

this study[50], **what these figures clearly and undoubtedly demonstrate is that these symbols, the same ones that we see in the Minoan script, were used as ideograms**.

We see how, at an early moment in the development of writing, the use of ideograms made it possible for the first time to share "concepts" between peoples who would maintain some kind of contact. In this example, it is seen that they could have adopted the same statuette as part of a common cult, but they would use the different words, in their own language, that their oral tradition sent them to refer to these linked ideas of mother, land, fertility, crops, etc.

Understanding the very obvious meaning of the figure, it seems very difficult for anyone to seriously maintain that, in reality, what is meant here are two syllables. For example, "/RAI-TJA/" as Gareth Owens's team would identify, forming a supposed word similar to Mycenaean Linear B, even though we are four or five centuries earlier in Turkmenistan or Pakistan, not on the island of Crete. Possible yes, but all readers must now agree that it seems very, very unlikely.

Several renowned researchers would then say that the fact of not being able to make a palaeographic analysis of this figure, due to the lack of a significant number of objects or elements, unfortunately implies abandoning all future hope of deciphering and understanding it. But this line of reasoning is, in the opinion of this modest author, completely unacceptable. And because of its simplicity this is a magnificent counterexample. As is the Altar of Malia.

By the way, the fact that we found in these figures these two symbols that we had already seen at the beginning of the work is an extraordinary validation of the meanings we gave them at the time, first within the context of the disc and then of the rest of the objects.

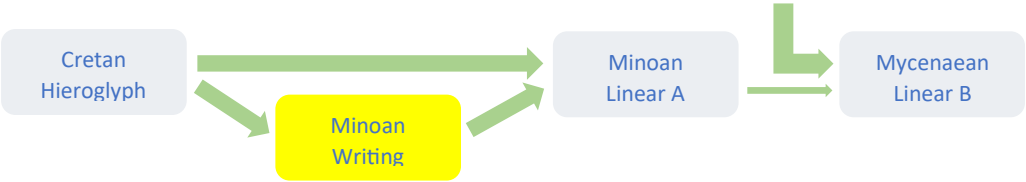
Finally, and to dispel any remaining doubt, we still have to refute the argument by which it is often said that, for an unknown script to be suspected of being of an ideographic type, it must present hundreds and even thousands of different symbols, plus their combinations. This author agrees in principle with this as long as this writing had reached a high level of development and implementation, thus encompassing all social order: administration, mercantile relations, laws, religion, popular uses, etc. In this way, it is evident that a multitude of symbols are needed to accommodate all the particular semantic elements that each field of use may need.

However, this requirement is not at all present in the case of the so-called Minoan script. The only archaeological artefacts found to date with symbols of this script are strictly limited to the religious sphere: the Phaistos Disc containing a communal prayer; the axes of Arkalochori which are votive objects with messages about the goddess; and the Altar of Malia that includes another prayer to the gods. To date, **there is no physical evidence that the Minoan script had any use on the island of Crete beyond the sacred precincts of the various Minoan palaces**. Even these votive figurines that appeared in Turkmenistan and Pakistan again show a single religious use.

In this sense, it is evident that 46 symbols cannot constitute such a rich ideographic basis as Chinese writing. But it is also evident that, and this work perfectly demonstrates it, by covering a series of basic concepts, these **46 symbols were good enough to generate elaborated and complex sentences** within this religious scope and during a specific period of time.

[50] A new world to explore. Link: (http://ensani.ir/file/download/article/1613995882-10369-99-53.pdf).

THE MINOAN WRITING HAS BEEN DECIPHERED



At this point it is understood that we have been able to decipher the Minoan writing by using the ideographic approach. As a result, we can now, some 4,000 years later, not only read but also fully understand all the inscriptions available today. Although in the future some minor changes to the meanings proposed in this document may be suggested.

Someone can always argue that the choice of meanings is subjective, and they will be right. As Gareth Owens said in his TED talk, unfortunately there is no Minoan-Rosetta stone. Therefore, any attempt to decipher the Minoan script, or any other, by means of the ideographic approach, will necessarily be based on subjective assumptions. This is by no means a problem *per se*. It will obviously depend first on the result, and then on whether it passes a general test of consistency with the context and with all known objects.

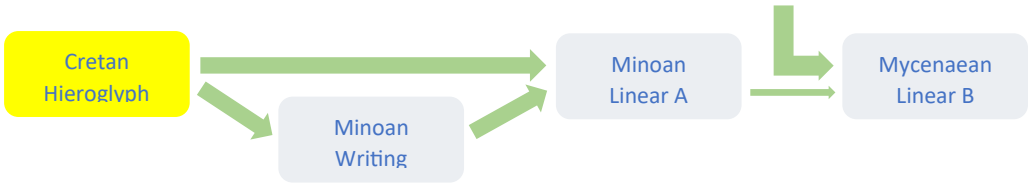
From the parallelism found in the texts of the Phaistos Disc, the Axe of Arkalochori, and the Altar of Malia, it is evident that at least the palaces of these three locations shared the knowledge of this writing. Nevertheless, it is very curious that there are only 3 or 4 objects with different materials: clay in Phaistos, the metal of the labrys in Arkalochori, or a stone altar in Malia. When it would have been reasonable to find many more remains, e.g. in clay, in all of them and in some other important location. **It is possible that these remains do exist and are confused among all the objects now catalogued as CHIC.**

Looking at it from another point of view, everything seems to indicate that this Minoan writing was known to a few people related to religious power, we suppose close to political power and, by the means used especially in Phaistos (very elaborate stamps), without the imminent intention of extending it to the rest of the population as a method of communication. On the contrary, it is most likely that at least for a time it was considered a secret knowledge of a magical nature, which would guarantee on the one hand the supposed effectiveness of the prayers, and on the other hand the very continuity of the religious caste. We can only speculate about what it was that in the end caused the transition to Linear A. Whether it was a smooth transition, or if, on the contrary, cataclysms such as that of the island of Santorini also had to do with this process of impulse, of the need for a common writing, perhaps also favoured by the probable befallen discredit of that religious caste.

In any case, we cannot fail to mention here that a great human and artistic figure must have existed at that time, who surely lived in the palace of Phaistos, and who was responsible for the design and creation of an object as spectacular and unique as his/her famous disc.

In conclusion, we have made a complete, coherent and serious study on the disc and the rest of the known Minoan inscriptions. Thus, and according to the aforementioned scientific method where, for example, a physical law that describes a phenomenon has a temporal validity only until a new one appears that expands or improves it, **we demand that this interpretation and translation of the Minoan script be taken into account and accepted, until another version may appear in the future and there is a consensus that considers it to be better than the present one**.

ABOUT THE CRETAN HIEROGLYPH



As we have been suggesting in the work, the so-called Minoan script was a creation and had its temporal development in the context of the religious precincts of the Minoan palaces, outside the Cretan Hieroglyph. To build it, they took as part of the symbols that probably existed a few centuries ago, and that can be seen in the numerous objects that have been found scattered on the island (more in its eastern half) containing these inscriptions, which are known as the body of the Cretan Hieroglyphic writing (CHIC). As its name suggests, it is considered that these ancient elements must have been hieroglyphics, that is, they should be pictograms/ideograms. The fact that, as we have seen, the Minoan script can be ideographic reinforces the idea that its original elements are also ideographic, so we believe that it is worth approaching its decipherment from this point of view.

Overall, looking over the collection of CHIC objects[51] it is easy to find numerous similarities, or outright coincidences, between their inscriptions and the Minoan symbols we have seen in this study. That is, in many cases the Minoan symbols do not present significant evolutions or modifications with respect to the original symbols of the Cretan Hieroglyph. That case we saw of the "CHIC_H_81 (Malia)" was a good example of this. But there are many more objects with this type of inscription that we can read and understand completely or partially at a simple glance.

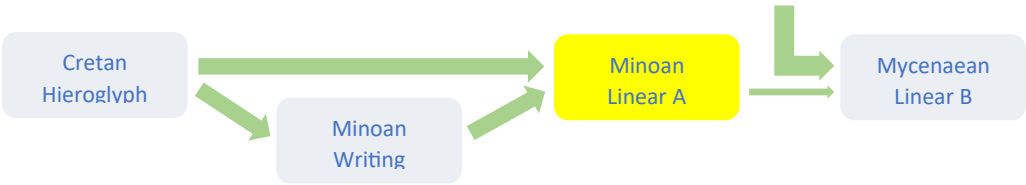
It is possible, as stated before, that **some of these objects actually belong to the Minoan script and are not well identified**. For example, the Altar of Malia was once classified as the object "CHIC 328 (Malia)".

It must also be assumed that the use and development of Cretan hieroglyphic symbols may in turn have evolved in parallel to the Minoan script. Among the wide variety of CHIC objects, very marked styles can be appreciated, for example, according to the physical origin of the pieces. Even within the same provenance, there are also obvious variations in style that could be caused by belonging to different eras or could even be related to different uses.

It is a very reasonable hypothesis to suppose that the meaning of the 46 known symbols of the Minoan script is the same as that which they had in the realm of the Cretan Hieroglyph. This is a great advantage to face the arduous task of advancing in the understanding of this scripture. Therefore, the great difficulty would be on the one hand in **identifying and trying to decipher those symbols that might not have been adopted in the Minoan script, and on the other hand in identifying the pictograms/ideograms already known in their different forms as a result of evolution and especially local variations**.

[51] On the website (https:// mwenge.github.io/linear0.xyz/) there is a graphic tool, programmed by Robert (Rob) Hogan, very useful for exploring CHIC objects, with all kinds of cross-references, possible translations (syllabic), etc. All illustrations are copyrighted by the École française d'Athènes.

ABOUT THE MINOAN LINEAR A



It has already been commented that it was very foreseeable that the Minoan writing present especially in the Phaistos Disc would not have a continuity or a deployment among the population, mainly due to the difficulty of having the stamps available. That was probably never the intention. The fact is that, apart from the disc and the fragment HM_992, no other remains of Minoan writing are known with these symbols made with stamps. On the other hand, the use of materials other than clay such as stone (Altar of Malia) or metal (Axes of Arkalochori) forced the use in parallel of symbols with a simpler design than those of the disc.

It is very likely that a second and deeper process of simplification came when this writing finally made the leap to a greater number of uses and users outside the religious sphere and was mixed again with the rest of the Cretan symbols. At that time the main challenge was undoubtedly to establish a **new standard in symbols to be able to be printed on clay, wood or even stone, using as few strokes as possible** made either with a marker (clay) or with a chisel (wood or stone). The fact that complicated resources were no longer needed to manufacture and distribute such as stamps, allowed them to be learned and used by a growing number of people.

The opening to new uses also brought the need to add **new symbols** to represent new concepts or objects (animals, goods, places), as well as to include **numerical characters**, and perhaps **new formal symbols** such as the beginning or end of sentences. All these elements progressively gave rise to the appearance and establishment of the Minoan Linear A language.

As a historical note, it must be said that due to the maritime-commercial relations of this people, this Linear A language was quite widespread in the eastern Mediterranean from about 1,700 to 1,400 B.C., when the Mycenaean civilization became predominant in the area, replaced the Minoans, and its syllabic writing system, Mycenaean Linear B, was gradually imposed for a period of 2 or 3 centuries before also succumbing[52].

The Mycenaean civilization has its origin in the Achaean people, of Indo-European origin who are believed to have migrated and settled in the Peloponnese peninsula around 1,600 BC. They also carried out intense commercial activity and rivalled the rest of the peoples around them, in particular the Minoans of Crete, whom they eventually subdued. Due to its geographical location and temporal development, it is very reasonable to think that its own script, Linear Mycenaean B, had some influence from the Linear Minoan A script, so that some of the symbols of Linear A that were widely used in the area ended up being introduced, with some small modifications, in Linear B. This is the only relationship that this author can establish between the Minoan Linear A and Mycenaean Linear B scripts. This relationship does not seem to us in any case to be of sufficient weight to justify that the Minoan script can be of a syllabic type just because it is an

[52] A curious fact is that the subsequent decline of the Mycenaean civilization occurred in a period where many other similar relevant changes also occurred throughout the eastern Mediterranean and western Asia. Some authors have called this period the "collapse of the Bronze Age". There are many references to this topic, see for example in Wikipedia: (https://es.wikipedia.org/wiki/Colapso_de_la_Edad_del_Bronce_Final).

ancient predecessor (not a precursor) on the island of Crete of Linear B, which is the fundamental argument of the followers of the syllabic approach to try to decipher the Minoan script.

Leaving Linear B and going back a few steps, we see that the evolution of the "spelling", in particular of the Minoan symbols, towards Linear A was not very homogeneous. For the version that would eventually prevail, they opted for forms more similar to those that appear on the Altar of Malia ("heaven", "earth", "govern", "multiply", "sprout", etc.), but they also chose some forms that had been seen in Axe of Arkalochori ("goddess", "holy will") and even others that we saw on the disc ("find", etc.).

Also, within Linear Minoan A there is an evolution and simplification of the designs over time, so that at first the symbols that came from the Minoan script are better recognized. This would be the case, for example, of the two axes of Arkalochori that we saw with supposed Linear A inscriptions, which we were able to identify as symbols of the Minoan script.

Currently, in the academic world there is still debate about whether the Linear A script is syllabic or not. It seems that after all these decades since the decipherment of Linear B in 1952 there is still no conclusive evidence in favour of the syllabic character of Linear A, so a part of the researchers no longer support this possibility, or let's say that they are beginning to be more open to other possibilities.

This work, in an effort to look at all of these writings in a holistic way, suggests that the Minoan Linear A script[53] was the natural evolution of the Cretan Hieroglyph together with the Minoan script, and that it would maintain its ideographic character at least in its beginnings. Thus again, as with the Cretan Hieroglyph, we believe that **it could be advanced in its decipherment by resolving or identifying the pictograms/ideograms in their different forms as a result of evolution or local variations.**

But this issue of advancing in the decipherment of both the Cretan Hieroglyph and Linear A is already far from the capabilities and intentions of this author.

[53] Ester Salgarella and Simon Castellan (Classics, St John's College, Cambridge) have developed a very useful website called SigLA to study inscriptions in Linear Minoan A (https://sigla.phis.me/index.html). There is also a similar graphical tool developed by Robert (Rob) Hogan to explore inscriptions in Linear A, link: (https://lineara.xyz/), as well as a similar one to explore inscriptions in Linear B, link: (https://linearb.xyz/).

APPENDICES

APPENDIX A. DISCUSSION ON THE NEW MINOAN SYMBOL

When it came to identifying the different symbols that appeared to us when reading the Phaistos Disc, we had some doubts when it came to distinguishing some of them. For example, with the "prisoner" symbol, which appears only in the middle of the phrase A27, on the outer end of the disc on its A-side and which is quite damaged. At first, we thought that it could be an impression of the stamp of the symbol "kid", which also appears only in the sentence B28, to which some later manual modifications could have been made. Thinking about how to detect these possible modifications of symbols, or differentiate them definitively, we simply came up with the idea of making some measurements of the size of the different marks of the symbols on the disk and comparing their sizes. The idea seemed simple: if two similar symbols were of a different size, then they would clearly be different stamps and therefore different symbols.

Since we do not have the original object to make measurements, we had to settle for taking measurements on the computer screen, on a still photograph, with a certain equal resolution for all measurements, and thus take relative data between the objects. In this case it was a little more complicated because the symbols were on different sides of the disk, then the photographs were different and the size of the disk in the two photographs had to be squared by eye so that they were comparable. Also considering that the disc is not perfectly round, the easiest thing to do was to adjust according to the sizes of the metal support that holds the disc.

In this way we measured a height of the "prisoner" symbol of 2.5 cm, and we could not establish the width because it has the "shield" symbol superimposed on one of its sides. Whereas the height of the "kid" was only 2.0 cm, and a maximum width of 1.1 cm. This implies that, regardless of the size of the photos we have on the computer screen, the height of the prisoner is 25% greater in relative terms than the height of the kid. This difference is much greater than any possible error of fine adjustment of sizes between the two photographs, so in this case it is very evident that they are different stamps and therefore, in accordance with the current consensus, they are considered to be different symbols.

[54]

However, the same problem arose between the symbols "2-legs" and "3-legs", as the latter is very similar to the former and appears only once on the disc, in the phrase B27 on the side B. On the other hand, the symbol "2-legs" appears several times on this same side B, for example in the phrase B22 or B30. In this case, measurements were in principle easier because only one photograph had to be used. Then, we first compared the measurements of all instances of the "2-legs" symbol and found that all of them had a height of 1.8 cm and a maximum width between the base of their legs of 1.0 cm. To measure the height of what we thought could be a new symbol, "3-legs", we had a problem because the image, the best we have found, shows a shadow at the base of the legs: if the shadow is taken into account as part of the figure, it would have a

[54] (left) Enlarged detail of the "kid" symbol in the photograph of the side B of the disc, in grid B28. (right) Enlarged detail of the "prisoner" symbol in the photograph of the side A of the disc, in grid A27 (both photographs were already referenced in the chapter of the Phaistos Disc).

height of 2.0 cm, but if it is not taken into account then it would have a height of 1.8 cm., which would be the same height as "2-legs". But the maximum width of the base of its legs (without considering the shadow or the possible third leg that appears on the right and that is not very defined in its lower part) is 1.2 cm, which represents a relative width at least 20% greater than that of the first, which would seem significant enough.

55

But unfortunately, in this specific case, the measurements obtained do not seem to give us a definitive criterion to discern whether they are two patterns or the same. Because someone might object that the case of the possible "3-legged" symbol would not be a new symbol but would actually be a one-off error or accident in printing the "2-legs" stamp on the fresh clay: the stamp was marked once and then there was a lateral displacement of the stamp to the left (if we look at the symbol from the front) and it was marked again on the clay on top of the previous one, leaving the right part as a shadow of the last marked[56].

However, we find quite a few objections to this possibility. It seems to us that if there had been a double marking, then the shadow of the left leg should also be seen, for example, and it is not visible. Nor does this hypothesis justify that the two main legs are at least 20% wider than the original figure. In addition, there are some differences in the drawing that cannot justify this hypothesis either, such as the different curvature that can be seen on the side of the trunk of the third leg (it would be assumed to be totally parallel to the trunk in the foreground), or the two points that now appear clearly in the centre of the new figure.

Thinking about the problem from another point of view, it is difficult to imagine that the person who executed the marking of the stamps on the clay did not want to correct such an obvious error and of which one is perfectly aware (if it were really a mistake) right at the end of the work, when solving it with fresh clay was very simple. And when it is evident that there were other rectifications on the disc: at least in the initial marks in the phrases A27 and A28. On the other hand, if we think that this person or persons belonged to the religious caste, the fact of leaving such a mistake reflected in a prayer to their gods would be something similar to committing a mortal sin, apart from the probable discredit that it would mean among their community.

For all these arguments we firmly believe that the misprint theory does not make any sense by itself, so **the symbol we have called "3-legs" is not a mistake, it comes from a different stamp and therefore is indeed a new Minoan symbol different from the "2-legs". And we found that this new symbol is not recognized in any of the numerous documents that we have reviewed to carry out this work**.

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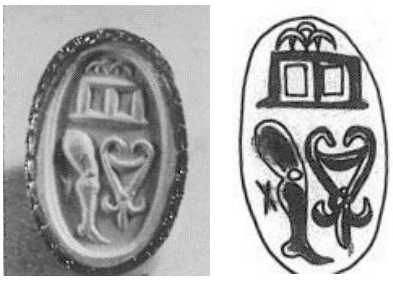
[55] (left) Enlarged detail of the "2-legs" symbol in the photograph of the B-side of the disc, in the B30 grid. (right) Enlarged detail of the "3-legged" symbol in the photograph of the B-side of the disc, in grid B27 (this photograph was already referenced in the chapter of the Phaistos Disc).

[56] In fact, a researcher recently raised this argument with us in a private conversation on the Academia platform. And here are reproduced the arguments with which this author answered.

APPENDIX B. REDISCOVERING AN ANCIENT STAR

Within the Cretan Hieroglyph there are numerous objects that show symbols in the shape of a human leg and also with two human legs, either in a very schematic form with simple straight lines (clay tablets), or in a very explicit form and with very elaborate drawings (seals). This led us to think that the symbol we have identified on the Phaistos Disc as No. 39 "2-legs" could actually represent a human trunk with its two legs. The meaning we assign to the symbol (future, prosperity) still works very well with this interpretation of the two legs instead of the two-legged support.

57 

58

In the same way, the symbol no. 45 "3-legs", whose design is just a small modification of the previous one, can also be interpreted as a representation of three human legs instead of a framework with three supporting legs. This interpretation is very interesting because the symbol of the three legs is very present throughout history, not only in this geographical area but also throughout Western Europe and has even reached the present day.

As a starting point, what we find very remarkable is that, unless mistaken, this symbol No. 45, different from No. 39, is not even identified to date by any document dealing with the Phaistos Disc or the Minoan script. Thus, we **would be the first to have identified this Minoan symbol that was used both in the Phaistos Disc and in the Axe of Arkalochori**.

And therefore, we are the first to suggest that this symbol No. 45 may actually be an **early representation of the symbol called "triskelion"**[59] **a superstar of antiquity, which would have been used by the Minoan civilization around 1,600 B.C**. (approximate dating that we suggest for the creation of the Phaistos Disc after the Santorini event). This is a date that, as we will see below, is much earlier than any other existing record. For this, we need to follow the trail of this symbol throughout history.

The term "triskelion" comes from the Greek word "triskele," which literally means three legs. This word "triskele" first referred to the triple spirals that proliferated during the Neolithic, e.g. the oldest dating being those of the island of Malta (approx. 4,400-3,600 BC), and some more recent ones such as the one at the entrance to the megalithic tomb of Newgrange, Ireland (approx. 3,200 BC).

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[57] CHIC 265 Kasteli S: CMS VI 94. Link: (https://mwenge.github.io/linear0.xyz/).
[58] CMS XI 251, Nationalmuseet, Kopenhage. Link: (https://arachne.dainst.org/entity/1160311/image/6899405).
[59] See an introduction to the "triskelion" symbol on Wikipedia. Link: (https://en.wikipedia.org/wiki/Triskelion).
[60] Schematic representation of a "triskelion" or "triskele" in its spiral version.

It is well documented that these triskeles designs were already used with numerous variations on the island of Crete from at least 1,850-1,700 BC (a period known in Cretan dating as "MM II"). See below the example of a seal, classified as belonging to the Cretan Hieroglyph, with an engraved image of a triskele with an inner triangle, and the result of imprinting it on a soft material such as clay.

61

This figure of the triskele was also known to the ancient Celtic people who probably migrated from Anatolia or the plains between the Black Sea and the Caspian Sea, and reached Europe (c. 1,200 BC) and present-day Iran and India. Triple spiral motifs will also be typical later in Mycenaean pottery.

But we have to jump to the VIII-VII centuries BC to begin to find explicit references to the symbol we are investigating. The "triskelion", in its version with legs bent at the knee, not with spirals, is known today above all for being a symbol rooted on the island of Sicily since it was a colony of Greece[62]. This island, shaped like a triangle with its three well-differentiated capes, adopted this symbol as its main emblem. Later, the head of a Gorgon or Medusa would be added in its central part to form what is still the coat of arms of its flag today. This figure is known as "Trinacria", which is believed to be a derivation of the term "Thrinakia" (which would come in turn from the word "thrinax", which means three peaks), which Homer used in his book Odyssey to name the island of Sicily (VIII century BC).

63

There are several theories about the origin of this "Trinacria"[64] but the most solid and accepted by the current consensus is that the original triskelion was brought by Minoan-Mycenaean merchants in their activity along the coasts of Sicily. This theory is based on the discovery in 1957 of a red ceramic plate ("dinos") at the Castellazzo di Palma site, on the south coast near Agrigento, Sicily, which showed a triskelion without the head of the Medusa (see photo below). It was dated to the end of the seventh century B.C. with what would be the oldest representation of this symbol found on this island and could be related to similar ceramic pieces found on the island of Crete that were dated to even earlier times. If it has, as we assume, a Minoan origin, then its meaning could be related to the desire or call for good fortune.

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[61] IDAI Object "Siegel CMS II,1 122" (Tholos, Agios Onouphrios). Link: (https://arachne.dainst.org/entity/1150728).

[62] There are many websites that talk about the symbol "Trinacria" e.g. at link: (https://www.ancient-symbols.com/symbols-directory/greek-triskelion.html).

[63] Flag of Sicily, with the "Trinacria".

[64] See the theories on the origin of the "Trinacria", e.g. at the following link: (https://www.giuseppegallo.design/design-and-communication/meaning-and-history-of-the-sicilian-triskele/).

65

On the other hand, the documentation also explains that this "triskelion" symbol would be one of the motifs that Greek soldiers, apparently in particular the Spartans (Lacedaemonian), wore painted on their shields. It is not known for sure whether the soldier chose a motif because he might like it in a particular way, perhaps because it represented some magical characteristic that he believed could benefit him, or if it was simply a way of intimidating the enemy, or if it was just an emblem of the family to which he belonged.

Among these motifs that were painted there was also the version of a single bent leg that, apparently, would have the meaning of strength (like our Minoan symbol no. 13 "mace"). You can even see some triskele motif not with three but with four legs, in what could be an early representation of the symbol later known as the "swastika".

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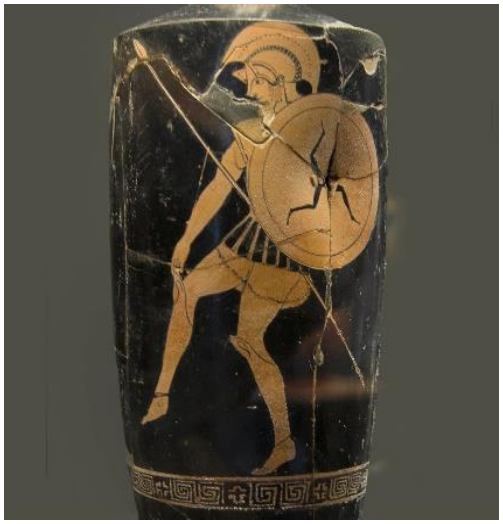
67

In the case of the three legs, based on the meaning we gave to this symbol on the Phaistos Disc, and assuming that the meaning was maintained over the centuries, the bearer of such a shield could be indicating his status as successful, fortunate, or, as they would say in ancient times, "blessed" by the Gods. Perhaps presumptuous, but surely very effective.

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[65] “Dinos” with triskelion, end VII century B.C., Archaeological Museum of Agrigento, photo © Giuseppe Gallo. Link: (https://www.giuseppegallo.design/design-and-communication/meaning-and-history-of-the-sicilian-triskele/).

[66] "Stamnos" from the sixth century B.C., which shows different decoration motifs on the shields of Greek soldiers. Among others, we see a four-legged triskele, the figure of a leg bent at the knee, just half of a drinking cup, and a bull's head with its two front legs. Photograph taken from the website: (https://es.topwar.ru/168207-antichnaja-keramika-i-oruzhie.html).

[67] Warrior with shield showing the triskelion symbol (ca. 470 B.C.), Archaeological Museum of Siracusa. Link: (https://www.giuseppegallo.design/design-and-communication/meaning-and-history-of-the-sicilian-triskele/).

Below are some photographs of beautiful vases, dated just before the classical Greece period, decorated with war scenes representing none other than the hero Achilles showing this symbol on his shield, which seems to us to be an extraordinary new validation of our interpretation of symbols.

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At least one of the scenes shown in the photographs refers to the Trojan War, which according to some scholars may have occurred between 1,194-1,184 B.C.[70] and which came down to us again through Homer's account in his Iliad (about 762 B.C.)[71]. We cannot know if oral tradition attributed this particular shield to this particular hero, so that the image would be a more or less "documented" representation or if, in the worst case, it was only a symbolic license taken by the artist who painted these scenes in the sixth century B.C.

Moving forward in time, between the sixth and fourth centuries B.C., throughout Asia Minor it was very common to see the figures of the triskeles and the Trinacria on coins. Examples: Aspendo (Pamphylia, eastern Mediterranean), Berrito and Thebe (Troas, territory around the city of Troy, between Scamander and Hellespont), Olba (Cilicia, between Armenia and Syria) and in coastal cities of the Lycian League (southwestern Anatolia). These models would also later be used by the Roman Empire on their coins.

72

[68] "Achilles dragging Hector's body" (520-510 B.C.). Theoi Project © Copyright 2000 - 2017 Aaron J. Atsma, Netherlands & New Zealand. Link: (https://www.theoi.com/Gallery/P21.3.html).

[69] "Achilles". Picture taken from web: (https://www.tumblr.com/chotomy/632737981546037248/throwback-to-that-time-my-ancient-greek-art).

[70] See for example the website (https://www.nationalgeographic.es/historia/guerra-troya-mitos-realidades).

[71] Some researchers have recently dated the publication of the Iliad to around 762 B.C. (eighth century B.C.) (https://latunicadeneso.wordpress.com/2013/03/01/una-investigacion-reciente-confirma-que-la-iliada-fue-escrita-en-el-siglo-viii-a-c/).

[72] (Left) Coin with triskele, Lycia (390-375 BC); (Centre) "Sirakous", Syracuse, with a triskelion identical to the famous plate found near Agrigento (ca. 336 BC); (Right) Coin of the Republic of Rome with the Trinacria (ca. 49 BC).

From the fourth century B.C. onwards, the spread of a new motif or symbol can be seen, a triangular figure formed by intertwined bows or arches, which will be embodied, for example, in the ceramics of Anatolia and Persia, but also in coins in the Germanic peoples or in the Lycian League, and even in runestones in Scandinavia. It is the figure known today as "Triquetra", a term that is derived from the adjective "triquetrus", which literally means three points, and which is just the Latinized version of the ancient Greek term "triskele". From the first centuries A.D. this term "Triquetra" will be used to refer to both this new figure and the old "triskele" or "triskelion".

73

This figure of the triquetra will be characteristic of the Celtic culture (in fact it is also known as the "Celtic knot") and will extend throughout the post-Roman era with a maximum diffusion between the seventh and eighth centuries A.D., especially in England and Ireland, the latter country where it will maintain its influence until the twelfth century. It will be used recurrently as a decorative element, both architecturally and accompanying written texts either on paper or chiselled. Its meaning has always been associated with all kinds of triads, from pagan to Christian religious circles. Sharing the same name, this meaning is also transferred to triskelion, which represents a radical change from its original meaning.

Probably related to the Christianised Celtic triquetra, there is a very popular tradition in Ireland whereby St. Patrick (c. 385-461 A.D.) would have used three-leaf clovers to teach the people the concept of the Holy Trinity. And in relation to this, there is also another tradition rooted in many cultures about considering the four-leaf clover as a sign or good luck charm, because statistically it is a variety that is difficult to find. The curious thing is that this interpretation somehow recovers the original meaning we gave to the Minoan symbol of the "3-legs" (lucky, successful).

But to explicitly see a triskelion again we have to go much further in time, when the Normans arrived on the island of Sicily in 1072 A.D. and exported the Trinacria symbol back to the Isle of Man, located between England and Ireland. Over time, this image was adopted there, which would replace its old symbol, a ship, which apparently had a Scandinavian origin. It has also survived to the present day in the form of a triskelion with legs lined with armour as the shield of its flag.

Already in the Middle Ages (thirteenth century A.D.) we find another curious triskelion decorating the ceiling of the church of Karja, in the town of Linnaka, Estonia, which tradition now says would symbolize the Christian Holy Trinity.

74

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[73] Schematic representation of a basic Triquetra. Consult website (https://en.wikipedia.org/wiki/Triquetra).
[74] Flag of the Isle of Man.
[75] Depiction of the triskelion from the church of Karja, Linnaka (Estonia), dated in the end of the thirteenth century A.D. See for example on the web (https://en.wikipedia.org/wiki/Karja_Church).

From the Middle Ages to the present day, we can trace the triskelion also in some coats of arms in Italy, Germany and England.

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These concepts and images have not been lost in time, but even today triskeles are still used as emblems of companies or institutions, such as the one used by the United States Department of Transportation, or the Irish Air Force.

Finally, it is worth mentioning a curious case that we have found. It is a seal of the Cretan Hieroglyph, therefore contemporary or perhaps older than the Phaistos Disc, similar to the one we saw at the beginning of this appendix, and which theoretically could represent a complex triskelion: where one of its three elements are the two human legs, and the other two are the head and two front legs of a bull (we had already seen this last symbol will also be used many centuries later in the shields of the Greek soldiers). The image of this curious stamp has been adopted by the University of Heidelberg as the logo for its research and documentation group "Corpus der Minoischen und Mykenischen Siegel" (CMS).

80

Summing up, we have tried here to briefly trace the path that the triskelion symbol and its relatives have had throughout history; and we have seen how the documentation found on the web pointed to a probable origin on the island of Crete. And on the other hand, we had shown in this work how the Minoan civilization already managed this concept and used it as another symbol of their writing present on the Phaistos Disc and on the Axe of Arkalochori, many centuries earlier than current sources recognize its first usage.

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[76] Unidentified coat of arms that appears e.g. in the link: (https://www.cssstrinakria.eu/tringlis.htm). This image, which came to us by chance as part of the business card of an Italian restaurant in Madrid, was actually the origin of this appendix.

[77] Coat of Arms Cossa family (or Coscia, or Cossavella), Naples, Italy: "Dizionario storico blasonico delle famiglie nobili e notabili italiane estinte e fiorenti" vol.II, pag. 329, compiled by Comm. G.B. di Crollalanza, published by the editors of the heraldic journal, Pisa 1,888. Link: (https://www.heraldrysinstitute.com/lang/es/coat_of_arms_card/idit/19216/Escudo+de+la+familia+Cossa).

[78] Coat of Arms/Seal of the City of Füsen (Ostallgäu from 1979), Bayern, Germany. They kept this image on their seal since 1,295 A.D. Link: (https://www.heraldry-wiki.com//heraldrywiki/index.php?title=F%C3%BCssen).

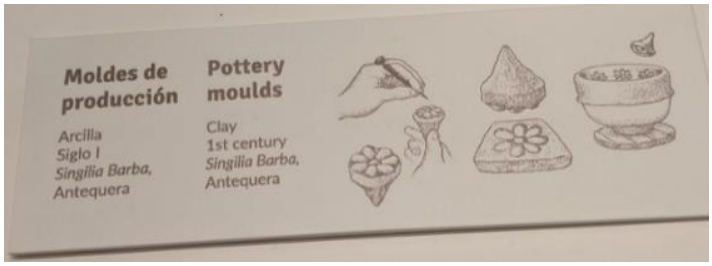
[79] Coat of arms of the Tremaine (or Tremayne) family, England, showing a triskelion made up of arms and fists. Link: (https://www.heraldrysinstitute.com/lang/es/coat_of_arms_card/idit/56617/Escudo+de+la+familia+Tremaine%2C+Tremayne).

[80] IDAI Object "Siegel CMS XI 336". Link: (https://arachne.dainst.org/entity/1160407).

APPENDIX C. MORE CERAMICS AND THE VENUS OF BENAOJÁN

On a recent visit to the Museum of the City of Malaga (Spain), on its top floor dedicated to a spectacular and very large sample of local archaeology, we found some ceramic pieces made with clay where again the use of stamps to print figures repetitively is evident. The use of these figures or symbols, as with the ceramic artifacts found at Phaistos, seems to us to be of a merely decorative character. The placard of these pieces indicated that they came from the Roman site called Singilia Barba, in the current town of Antequera, province of Malaga, on the southwest coast of Spain. And they are dated to the first century A.D., that is, about 1,600 years after the Minoans used this technique on the island of Crete.

We find it very surprising to find these apparently very specific examples of the use of stamps in such a distant location, both in space and time, when it seemed that this artistic practice was completely discontinued after the end of the Minoan civilization. It is like finding in the past a faint echo of more distant times. It is also striking that one of the several decorative motifs that can be seen in these pieces is very similar to the Minoan symbol that we call "flower", although here it is shown with only 7 petals, and not 8 as it had in the Phaistos Disc.

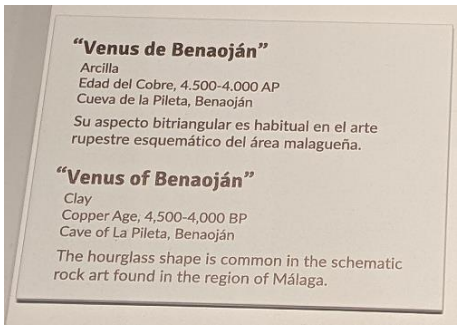


[81]

On the other hand, we also saw a clay figurine with a female shape that could not help but remind us of those found in the locations of Altyn-Depe (Turkmenistan) and Gandhara (Pakistan). It is presented as the "Venus of Benaoján", due to the location where it was found, in the so-called "Cueva de la Pileta", in Benaoján, interior of the province of Malaga (Spain).

[81] Photographs of several clay utensils dated to the first century A.D., which could well be a kind of plate and a vessel, as well as its placard showing the use of stamps to mark decorative figures or symbols. It can be seen that both have a hole in the centre of the piece, which could indicate a use related to the filtering of water or liquids, in a similar way to current containers for plants or flowers. Photographs taken at the Museum of the City of Malaga by the author himself.

The placard says that it is dated approximately between 4,500-4,000 BP, a nomenclature that is typical in archaeological objects for which Carbon-14 dating has been used. "BP" stands for "Before the Present", with the "present" being the year 1,950 A.D [82]. Then the range 4,500-4,000 BP corresponds, ignoring some small statistical adjustment, to the period 2,550-2,050 B.C., which brings us to the end of the Copper Age or beginning of the Bronze Age. This implies that this object would be somewhat earlier than the statuettes found in Altyn-Depe and Gandhara.



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The resemblance to these is evident. The representation of female sexual attributes is obvious. In particular, the representation of the pubis by means of a series of marks or grouped holes seems very relevant, an identical image although less elaborate to that in the other figures in which the representation of the symbol that would later appear in the Phaistos Disc and that in this work we have called "triangle" was more clearly appreciated, related to fertility and agricultural production. In addition, there are holes in the upper part that would indicate that the piece could also have been used as a necklace or pendant, as well as some of the figures of Altyn-Depe or the figure found in Gandhara.

The silhouette of this figure is the typical one that we will see later, for example, in the copper ingots of the Bronze Age (means of payment), also known as "bull's skin" or, according to its shape term, "hourglass". Actually, these bull's skin ingots could be the conceptual origin of the Minoan symbol we called “hide” (wealth/riches). We see that the relationship between the means of payment and the deities is a constant over time: at first imitating their original form of representation, and then almost to the present day including their images on coins.

Although there is not much information about the people who lived in this place in that period, this new figure could be an older representation of the concept of Mother Earth, as it has many similarities with other known ones. In any case, it seems to show us an idea and a format that seems to be ubiquitous in many of the Indo-European cultures of the Copper and Bronze Ages, from one end to the other, which would have been in contact by inland routes or, more likely, through the Mediterranean Sea.

The relevant fact here is that **the future use of ideograms shared between the different peoples will serve in the first place to provide definition and enrich this type of cult objects**, which had already been present for several centuries.

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[82] You can see, for example, a reference to this nomenclature and the related method of dating on the web (https://archpress.lib.sfu.ca/index.php/archpress/catalog/download/41/13/571?inline=1).

[83] Photograph of the referred placard taken at the Museum of the City of Malaga by the author himself.

[84] Photograph of the clay statuette known as "Venus of Benaoján", Malaga (Spain), dated between 2,550-2,050 B.C. Taken from the website of the Archaeological Museum of Alicante, temporary exhibition "Idols, millenary looks", January-April 2020. Link: (https://www.marqalicante.com/exposiciones/Catalogo/16/venus-P915.html).

APPENDIX D. TABLE OF MINOAN SYMBOLS CORRESPONDENCE

Cretan Hieroglyph	Axe of Arkalochori	Phaistos Disc	Fragment HM_992	CHIC_H_81	Altar of Malia	Linear A
						[...]

Title	Decoding Minoan writing
Version	v3.0
Language	English
Main Changes	“About Minoan pottery in Phaistos” (disc), “Partial parallel texts” (axe) and “About Consensus“ (altar) new sections added, Conclusion chapter rewritten, and new Appendix A, B and C.
Author	Juan Antonio Lozano Mendoza
© Registered	July 15th 2024

Title	Descifrando la escritura Minoica
Version	v2.1
Language	Spanish
Main Changes	Translation into Spanish
Author	Juan Antonio Lozano Mendoza
© Registered	January 18th 2024

Title	Decoding Minoan writing
Version	v2.0
Language	English
Main Changes	Translation into English. “About Consensus” new sections, Conclusions 1&2 and Appendix (table of signs) added.
Author	Juan Antonio Lozano Mendoza
© Registered	January 12th 2024

Title	Descifrando la escritura Minoica
Version	v1.0
Language	Spanish
Main Changes	First version
Author	Juan Antonio Lozano Mendoza
© Registered	December 1st 2023